

s described in the theorem. Such a vector is called a **steady-state distribution vector**. We indicate the argument using a 3×3 matrix T . We know that the population distribution vector after m time periods is $T^m p$. If we let

$$p = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} \quad \text{and} \quad s = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix},$$

then Theorem 1.19 tells us that $T^m p$ is approximately

$$\begin{bmatrix} s_1 & s_1 & s_1 \\ s_2 & s_2 & s_2 \\ s_3 & s_3 & s_3 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} s_1 p_1 + s_1 p_2 + s_1 p_3 \\ s_2 p_1 + s_2 p_2 + s_2 p_3 \\ s_3 p_1 + s_3 p_2 + s_3 p_3 \end{bmatrix}.$$

Because $p_1 + p_2 + p_3 = 1$, this vector becomes

$$\begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix}.$$

Thus, after many time periods, the population distribution vector is approximately equal to the steady-state vector s for any choice of initial population distribution vector p .

There are two ways we can attempt to compute the steady-state vector s of a regular transition matrix T . If we have a computer handy, we can simply raise T to a sufficiently high power so that all column vectors are the same, as far as the computer can print them. The software LINTEK or MATLAB can be used to do this. Alternatively, we can use part (2) of Theorem 1.19 and solve for s in the equation

$$Ts = s. \quad (1)$$

In solving Eq. (1), we will be finding our first *eigenvector* in this text. We will have a lot more to say about eigenvectors in Chapter 5.

Using the identity matrix I , we can rewrite Eq. (1) as

$$\begin{aligned} Ts &= Is \\ Ts - Is &= 0 \\ (T - I)s &= 0. \end{aligned}$$

The last equation represents a homogeneous system of linear equations with coefficient matrix $(T - I)$ and column vector s of unknowns. From all the solutions of this homogeneous system, choose the solution vector with positive entries that add up to 1. Theorem 1.19 assures us that this solution exists and is unique. Of course, the homogeneous system can be solved easily using a computer. Either LINTEK or MATLAB will reduce the augmented matrix to a form from which the solutions can be determined easily. We illustrate both methods with examples.

EXAMPLE 6 Use the routine MATCOMP in LINTEK, and raise the transition matrix to powers to find the steady-state distribution vector for the Markov chain in Example 1, having states labeled

poor, middle income, rich.

SOLUTION Using MATCOMP and experimenting a bit with powers of the matrix T in Example 1, we find that

$$T^{60} \approx \begin{array}{ccc} \text{poor} & \text{mid} & \text{rich} \\ \begin{bmatrix} .3872054 & .3872054 & .3872054 \\ .4680135 & .4680135 & .4680135 \\ .1447811 & .1447811 & .1447811 \end{bmatrix} & \begin{array}{l} \text{poor} \\ \text{mid} \\ \text{rich} \end{array} \end{array}$$

Thus eventually about 38.7% of the population is poor, about 46.8% is middle income, and about 14.5% is rich, and these percentages no longer change as time progresses further over 20-year periods. ■

EXAMPLE 7 The inhabitants of a vegetarian-prone community agree on the following rules:

1. No one will eat meat two days in a row.
2. A person who eats no meat one day will flip a fair coin and eat meat on the next day if and only if a head appears.

Determine whether this Markov-chain situation is regular; and if so, find the steady-state distribution vector for the proportions of the population eating no meat and eating meat.

SOLUTION The transition matrix T is

$$T = \begin{array}{cc} \begin{array}{cc} \text{no meat} & \text{meat} \end{array} \\ \begin{bmatrix} \frac{1}{2} & 1 \\ \frac{1}{2} & 0 \end{bmatrix} & \begin{array}{l} \text{no meat} \\ \text{meat} \end{array} \end{array}$$

Because T^2 has no zero entries, the Markov chain is regular. We solve

$$(T - I)\mathbf{s} = \mathbf{0}, \quad \text{or} \quad \begin{bmatrix} -\frac{1}{2} & 1 \\ \frac{1}{2} & -1 \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

We reduce the augmented matrix:

$$\left[\begin{array}{cc|c} -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & -1 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -2 & 0 \\ \frac{1}{2} & -1 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array} \right].$$

Thus we have

$$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 2r \\ r \end{bmatrix} \quad \text{for some scalar } r.$$

But we must have $s_1 + s_2 = 1$, so $2r + r = 1$ and $r = \frac{1}{3}$. Consequently, the steady-state population distribution is given by the vector $\begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$. We see that, eventually, on each day about $\frac{2}{3}$ of the people eat no meat and the other $\frac{1}{3}$ eat meat. This is independent of the initial distribution of population between the states. All might eat meat the first day, or all might eat no meat; the steady-state vector remains $\begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$ in either case. ■

If we were to solve Example 7 by reducing an augmented matrix with a computer, we should add a new row corresponding to the condition $s_1 + s_2 = 1$ for the desired steady-state vector. Then the unique solution could be seen at once from the reduction of the augmented matrix. This can be done using pencil-and-paper computations just as well. If we insert this as the first condition on s and rework Example 7, our work appears as follows:

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & -1 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & \frac{3}{2} & \frac{1}{2} \\ 0 & -\frac{3}{2} & -\frac{1}{2} \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 \end{array} \right]$$

Again, we obtain the steady-state vector $\begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$.

SUMMARY

1. A transition matrix for a Markov chain is a square matrix with nonnegative entries such that the sum of the entries in each column is 1.
2. The entry in the i th row and j th column of a transition matrix is the proportion of the population in state j that moves to state i during one time period of the chain.
3. If the column vector $\mathbf{p}$ is the initial population distribution vector between states in a Markov chain with transition matrix T , the population distribution vector after one time period of the chain is $T\mathbf{p}$.
4. If T is the transition matrix for one time period of a Markov chain, then T^m is the transition matrix for m time periods.
5. A Markov chain and its associated transition matrix T are called *regular* if there exists an integer m such that T^m has no zero entries.
6. If T is a regular transition matrix for a Markov chain,
 - a. The columns of T^m all approach the same probability distribution vector $\mathbf{s}$ as m becomes large;
 - b. $\mathbf{s}$ is the unique probability distribution vector satisfying $T\mathbf{s} = \mathbf{s}$; and

- c. As the number of time periods increases, the population distribution vectors approach $\mathbf{s}$ regardless of the initial population distribution vector $\mathbf{p}$. Thus $\mathbf{s}$ is the steady-state population distribution vector.

EXERCISES

In Exercises 1–8, determine whether the given matrix is a transition matrix. If it is, determine whether it is regular.

1.
$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

2.
$$\begin{bmatrix} 0 & \frac{1}{4} \\ 1 & \frac{3}{4} \end{bmatrix}$$

3.
$$\begin{bmatrix} .2 & .1 & .3 \\ .4 & .5 & -.1 \\ .4 & .4 & .8 \end{bmatrix}$$

4.
$$\begin{bmatrix} .1 & .2 \\ .3 & .4 \\ .6 & .4 \end{bmatrix}$$

5.
$$\begin{bmatrix} .5 & 0 & 0 & .5 \\ .5 & .5 & 0 & 0 \\ 0 & .5 & .5 & 0 \\ 0 & 0 & .5 & .5 \end{bmatrix}$$

6.
$$\begin{bmatrix} .3 & .2 & 0 & .5 \\ .4 & .2 & 0 & .1 \\ .1 & .2 & 1 & .2 \\ .2 & .4 & 0 & .2 \end{bmatrix}$$

7.
$$\begin{bmatrix} 0 & .5 & .2 & .2 & .1 \\ .3 & 0 & .1 & .8 & .5 \\ 0 & 0 & .4 & 0 & .1 \\ .7 & .5 & .1 & 0 & .1 \\ 0 & 0 & .2 & 0 & .2 \end{bmatrix}$$

8.
$$\begin{bmatrix} 0 & .1 & 0 & 0 & .9 \\ 0 & .2 & 0 & 0 & .1 \\ 0 & .3 & 0 & 1 & 0 \\ 1 & .1 & 0 & 0 & 0 \\ 0 & .3 & 1 & 0 & 0 \end{bmatrix}$$

In Exercises 9–12, let $T = \begin{bmatrix} .3 & .7 & .4 \\ .4 & .2 & .1 \\ .3 & .1 & .5 \end{bmatrix}$ be the transition matrix for a Markov chain, and let $\mathbf{p} = \begin{bmatrix} .3 \\ .2 \\ .5 \end{bmatrix}$ be the initial population distribution vector.

- Find the proportion of the state 2 population that is in state 3 after two time periods.
- Find the proportion of the state 3 population that is in state 2 after two time periods.
- Find the proportion of the total population that is in state 3 after two time periods.
- Find the population distribution vector after two time periods.

In Exercises 13–18, determine whether the given transition matrix with the indicated distribution of zero entries and nonzero $\times$ entries is regular.

13.
$$\begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \end{bmatrix}$$

14.
$$\begin{bmatrix} \times & 0 & \times \\ 0 & 0 & \times \\ \times & \times & 0 \end{bmatrix}$$

15.
$$\begin{bmatrix} 0 & \times & 0 \\ \times & \times & \times \\ 0 & \times & 0 \end{bmatrix}$$

16.
$$\begin{bmatrix} \times & 0 & \times & \times \\ \times & \times & \times & \times \\ \times & 0 & \times & \times \\ \times & 0 & \times & \times \end{bmatrix}$$

17.
$$\begin{bmatrix} 0 & \times & \times & \times \\ \times & 0 & \times & \times \\ 0 & 0 & \times & \times \\ 0 & 0 & \times & \times \end{bmatrix}$$

18.
$$\begin{bmatrix} 0 & 0 & \times & \times \\ \times & 0 & 0 & \times \\ 0 & \times & 0 & \times \\ 0 & 0 & 0 & \times \end{bmatrix}$$

In Exercises 19–24, find the steady-state distribution vector for the given transition matrix of a Markov chain.

19.
$$\begin{bmatrix} \frac{2}{3} & 1 \\ \frac{1}{3} & 0 \end{bmatrix}$$

20.
$$\begin{bmatrix} \frac{3}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$

21.
$$\begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} \end{bmatrix}$$

22.
$$\begin{bmatrix} 1 & \frac{1}{4} & \frac{1}{2} \\ 0 & \frac{1}{4} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

23.
$$\begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{4} & 0 & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{3} \end{bmatrix}$$

24.
$$\begin{bmatrix} \frac{1}{5} & \frac{1}{5} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{5} & \frac{1}{3} \\ \frac{2}{5} & \frac{3}{5} & \frac{1}{3} \end{bmatrix}$$

25. Mark each of the following True or False.
- All entries in a transition matrix are nonnegative.
 - Every matrix whose entries are all nonnegative is a transition matrix.

- c. The sum of all the entries in an $n \times n$ transition matrix is 1.
 - d. The sum of all the entries in an $n \times n$ transition matrix is n .
 - e. If a transition matrix contains no zero entries, it is regular.
 - f. If a transition matrix is regular, it contains no nonzero entries.
 - g. Every power of a transition matrix is again a transition matrix.
 - h. If a transition matrix is regular, its square has equal column vectors.
 - i. If a transition matrix T is regular, there exists a unique vector s such that $Ts = s$.
 - j. If a transition matrix T is regular, there exists a unique population distribution vector s such that $Ts = s$.
26. Estimate A^{100} , if A is the matrix in Exercise 20.
27. Estimate A^{100} , if A is the matrix in Exercise 23.

Exercises 28–33 deal with the following Markov chain. We classify the women in a country according as to whether they live in an urban (U), suburban (S), or rural (R) area. Suppose that each woman has just one daughter, who in turn has just one daughter, and so on. Suppose further that the following are true:

For urban women, 10% of the daughters settle in rural areas, and 50% in suburban areas.

For suburban women, 20% of the daughters settle in rural areas, and 30% in urban areas.

For rural women, 20% of the daughters settle in the suburbs, and 70% in rural areas.

Let this Markov chain have as its period the time required to produce the next generation.

- 28. Give the transition matrix for this Markov chain, taking states in the order U, S, R.
- 29. Find the proportion of urban women whose granddaughters are suburban women.
- 30. Find the proportion of rural women whose granddaughters are also rural women.

31. If the initial population distribution vector

for all the women is $\begin{bmatrix} .4 \\ .5 \\ .1 \end{bmatrix}$, find the population

distribution vector for the next generation.

- 32. Repeat Exercise 31, but find the population distribution vector for the following (third) generation.
- 33. Show that this Markov chain is regular, and find the steady-state probability distribution vector.

Exercises 34–39 deal with a simple genetic model involving just two types of genes, G and g .

Suppose that a physical trait, such as eye color, is controlled by a pair of these genes, one inherited from each parent. A person may be classified as being in one of three states:

Dominant (type GG), Hybrid (type Gg),
Recessive (type gg).

We assume that the gene inherited from a parent is a random choice from the parent's two genes—that is, the gene inherited is just as likely to be one of the parent's two genes as to be the other. We form a Markov chain by starting with a population and always crossing with hybrids to produce offspring. We take the time required to produce a subsequent generation as the time period for the chain.

- 34. Give an intuitive argument in support of the idea that the transition matrix for this Markov chain is

$$T = \begin{array}{c|ccc} & D & H & R \\ \hline D & \frac{1}{2} & \frac{1}{4} & 0 \\ H & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ R & 0 & \frac{1}{4} & \frac{1}{2} \end{array}$$

- 35. What proportion of the third-generation offspring (after two time periods) of the recessive (gg) population is dominant (GG)?
- 36. What proportion of the third-generation offspring (after two time periods) of the hybrid (Gg) population is not hybrid?

37. If initially the entire population is hybrid, find the population distribution vector in the next generation.
38. If initially the population is evenly divided among the three states, find the population distribution vector in the third generation (after two time periods).
39. Show that this Markov chain is regular, and find the steady-state population distribution vector.
40. A state in a Markov chain is called **absorbing** if it is impossible to leave that state over the next time period. What characterizes the transition matrix of a Markov chain with an absorbing state? Can a Markov chain with an absorbing state be regular?
41. Consider the genetic model for Exercises 34–39. Suppose that, instead of always crossing with hybrids to produce offspring, we always cross with recessives. Give the transition matrix for this Markov chain, and show that there is an absorbing state. (See Exercise 40.)
42. A Markov chain is termed **absorbing** if it contains at least one absorbing state (see Exercise 40) and if it is possible to get from any state to an absorbing state in some number of time periods.
- Give an example of a transition matrix for a three-state absorbing Markov chain.
 - Give an example of a transition matrix for a three-state Markov chain that is not absorbing but has an absorbing state.
43. With reference to Exercise 42, consider an absorbing Markov chain with transition matrix T and a single absorbing state. Argue that, for any initial distribution vector $\mathbf{p}$, the vectors $T^n \mathbf{p}$ for large n approach the vector containing 1 in the component that corresponds to the absorbing state and zeros elsewhere. [SUGGESTION: Let m be such that it is possible to reach the absorbing state from any state in m time periods, and let q be the smallest entry in the row of T^m corresponding to the absorbing state. Form a new chain with just two states, Absorbing (A) and Free (F), which has as time period m time periods of the original chain, and with probability q of moving from state F to state

A in that time period. Argue that, by starting in any state in the original chain, you are more likely to reach an absorbing state in mr time periods than you are by starting from state F in the new chain and going for r time periods. Using the fact that large powers of a positive number less than 1 are almost 0, show that for the two-state chain, the population distribution vector approaches $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ as the number of time periods increases, regardless of the initial population distribution vector.]

44. Let T be an $n \times n$ transition matrix. Show that, if every row and every column have fewer than $n/2$ zero entries, the matrix is regular.

In Exercises 45–49, find the steady-state population distribution vector for the given transition matrix. See the comment following Example 7.

45. $\begin{bmatrix} 0 & \frac{1}{4} \\ 1 & \frac{3}{4} \end{bmatrix}$

46. $\begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{3}{4} \end{bmatrix}$

47. $\begin{bmatrix} \frac{1}{5} & 1 \\ \frac{4}{5} & 0 \end{bmatrix}$

48. $\begin{bmatrix} 0 & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{4} & 0 \end{bmatrix}$

49. $\begin{bmatrix} \frac{1}{5} & \frac{3}{4} & \frac{1}{8} \\ \frac{4}{5} & 0 & \frac{3}{8} \\ 0 & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$



In Exercises 50–54, find the steady-state population distribution vector by (a) raising the matrix to a power and (b) solving a linear system. Use LINTEK or MATLAB.

50. $\begin{bmatrix} .1 & .3 & .4 \\ .2 & 0 & .2 \\ .7 & .7 & .4 \end{bmatrix}$

51. $\begin{bmatrix} .3 & .3 & .1 \\ .1 & .3 & .5 \\ .6 & .4 & .4 \end{bmatrix}$

52. $\begin{bmatrix} .1 & 0 & .2 & .5 \\ .4 & 0 & .2 & .5 \\ .2 & .5 & .4 & 0 \\ .3 & .5 & .2 & 0 \end{bmatrix}$

53. $\begin{bmatrix} .5 & 0 & 0 & 0 & .9 \\ .5 & .4 & 0 & 0 & 0 \\ 0 & .6 & .3 & 0 & 0 \\ 0 & 0 & .7 & .2 & 0 \\ 0 & 0 & 0 & .8 & .1 \end{bmatrix}$

54. The matrix in Exercise 8

1.8

APPLICATION TO BINARY LINEAR CODES (OPTIONAL)

We are not concerned here with secret codes. Rather, we discuss the problem of encoding information for transmission so that errors occurring during transmission or reception have a good chance of being detected, and perhaps even being corrected, by an appropriate decoding procedure. The diagram

message $\rightarrow$ encode $\rightarrow$ transmit $\rightarrow$ receive $\rightarrow$ decode $\rightarrow$ message

shows the steps with which we are concerned. Errors could be caused at any stage of the process by equipment malfunction, human error, lightning, sunspots, cross talk interference, etc.

Numerical Representation of Information

All information can be reduced to sequences of numbers. For example, we could number the letters of our alphabet and represent every word in our language as a finite sequence of numbers. We concentrate on how to encode numbers to detect and possibly correct errors.

We are accustomed to expressing numbers in decimal (base 10) notation, using as *alphabet* the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. However, they also can be expressed using any integer base greater than or equal to 2. A computer works in binary (base 2) notation, which uses the smaller alphabet $\{0, 1\}$; the number 1 can be represented by the presence of an electric charge or by current flowing, whereas the absence of a charge or current can represent 0. The American National Standard Code for Information Interchange (ASCII) has 256 characters and is widely used in personal computers. It includes all the characters that we customarily find on typewriter keyboards, such as

*A a B b Z z 1 2 3 4 5 6 7 8 9 0 , ; ? * & # ! + - / ' " .*

The 256 characters are assigned numbers from 0 to 255. For example, *S* is assigned the number 83 (decimal), which is 1010011 (binary) because, reading 1010011 from left to right, we see that

$$1(2^6) + 0(2^5) + 1(2^4) + 0(2^3) + 0(2^2) + 1(2^1) + 1(2^0) = 83.$$

The ASCII code number for the character 7 is 55 (decimal), which is 110111 (binary). Because $2^8 = 256$, each character in the ASCII code can be represented by a sequence of eight 0's or 1's; the *S* is represented by 01010011 and 7 by 00110111. This discussion makes it clear that all information can be encoded using just the *binary alphabet* $\mathbb{B} = \{0, 1\}$.

Message Words and Code Words

An algebraist refers to a sequence of characters from some alphabet, such as 01010011 using the alphabet $\mathbb{B}$ or the sequence of letters *glypt* using our usual letter alphabet, as a *word*; the computer scientist refers to this as a *string*. As we

discuss encoding words so that transmission errors can be detected, it is convenient to use as small an alphabet as possible, so we restrict ourselves to the **binary words** using the alphabet $\mathbb{B} = \{0, 1\}$. Rather than give illustrations using words of eight characters as in the ASCII code, let us use words of just four characters; the 16 possible words are

0000 0001 0010 0011 0100 0101 0110 0111
1000 1001 1010 1011 1100 1101 1110 1111.

An error in transmitting a word occurs when a 1 is changed to a 0 or vice versa during transmission. The first two illustrations exhibit an inefficient way and an efficient way of detecting a transmission error in which *only one* erroneous interchange of the characters 0 and 1 occurs. In each illustration, a binary *message word* is encoded to form the *code word* to be transmitted.

ILLUSTRATION 1 Suppose we wish to send 1011 as the binary message word. To detect any single-error transmission, we could send each character twice—that is, we could *encode* 1011 as 11001111 when we send it. If a single error is made in transmission of the code word 11001111 and the recipient knows the encoding scheme, then the error will be detected. For illustration, if the received code word is 11001011, the recipient knows there is an error because the fifth and sixth characters are different. Of course, the recipient does not know whether 0 or 1 is the correct character. But note that not all double-error transmissions can be detected. For example, if the received code word is 11000011, the recipient perceives no error, and obtains 1001 upon decoding, which was not the message sent. ■

One problem with encoding a word by repeating every character as in Illustration 1 is that the code word is twice as long as the original message word. There is a lot of *redundancy*. The next illustration shows how we can more efficiently achieve the goal of warning the receiver whenever a single error has been committed.

ILLUSTRATION 2 Suppose again that we wish to transmit a four-character word on the alphabet $\mathbb{B}$. Let us denote the word symbolically by $x_1x_2x_3x_4$, where each x_i is either 0 or 1. We make use of *modulo 2 arithmetic* on $\mathbb{B}$, where

$$0 + 0 = 0, \quad 1 + 0 = 0 + 1 = 1, \quad \text{and} \quad 1 + 1 = 0 \quad (\text{modulo 2 sums})$$

and where subtraction is the same as addition, so that $0 - 1 = 0 + 1 = 1$. Multiplication is as usual: $1 \cdot 0 = 0 \cdot 1 = 0$ and $1 \cdot 1 = 1$. We append to the word $x_1x_2x_3x_4$ the modulo 2 sum

$$x_5 = x_1 + x_2 + x_3 + x_4. \quad (1)$$

This amounts to appending the character 0 if the message word contains an even number of characters 1, and appending a 1 if the number of 1's in the

word is odd. Note that the result is a five-character code word $x_1x_2x_3x_4x_5$ definitely containing an even number of 1's. Thus we have

$$x_1 + x_2 + x_3 + x_4 + x_5 = 0 \pmod{2}.$$

If the message word is 1011, as in Illustration 1, then the encoded word is 10111. If a single error is made in transmitting this code word, changing a single 0 to 1 or a single 1 to 0, then the modulo 2 sum of the five characters will be 1 rather than 0, and the recipient will recognize that there has been an error in transmitting the code word. ■

Illustration 2 attained the goal of recognizing single-error transmissions with less redundancy than in Illustration 1. In Illustration 2, we used just one extra character, whereas in Illustration 1 we used four extra characters. However, using the scheme in Illustration 1, we were able to identify which character of the message was affected, whereas the technique in Illustration 2 showed only that at least one error had been made.

Computers use the scheme in Illustration 2 when storing the ASCII code number of one of the 256 ASCII characters. An extra 0 or 1 is appended to the binary form of the code number, so that the number of 1's in the augmented word is even. When the encoded ASCII character is retrieved, the computer checks that the number of 1's is indeed even. If it is not, it can try to read that binary word again. The user may be warned that there is a PARITY CHECK problem if the computer is not successful.

Terminology

Equation 1 in Illustration 2 is known as a **parity-check equation**. In general, starting with a **message word** $x_1x_2 \cdots x_k$ of k characters, we encode it as a **code word** $x_1x_2 \cdots x_k \cdots x_n$ of n characters. The first k characters are the **information portion** of the encoded word, and the final $n - k$ characters are the **redundancy portion** or **parity-check portion**.

We introduce more notation and terminology to make our discussion easier. Let $\mathbb{B}^n$ be the set of all binary words of n consecutive 0's or 1's. A **binary code** C is any subset of $\mathbb{B}^n$. We can identify a vector of n components with each word in $\mathbb{B}^n$ —namely, the vector whose i th component is the i th character in the word. For example, we can identify the word 1101 with the row vector $[1, 1, 0, 1]$. It is convenient to denote the set of all of these row vectors with n components by $\mathbb{B}^n$ also. This notation is similar to the notation $\mathbb{R}^n$ for all n -component vectors of real numbers. On occasion, we may find it convenient to use column vectors rather than row vectors.

The **length** of a word u in $\mathbb{B}^n$ is n , the number of its components. The **Hamming weight** $\text{wt}(u)$ of u is the number of components that are 1. Given two binary words u and v in $\mathbb{B}^n$, the **distance between them**, denoted by $d(u, v)$, is the number of components in which the entries in u and v are different, so that one of the words has a 0 where the other has a 1.

ILLUSTRATION 3 Consider the binary words $u = 11010011$ and $v = 01110111$. Both words have length 8. Also, $\text{wt}(u) = 5$, whereas $\text{wt}(v) = 6$. The associated vectors differ in the first, third, and sixth components, so $d(u, v) = 3$. ■

We can define addition on the set $\mathbb{B}^n$ by adding modulo 2 the characters in the corresponding positions. Remembering that $1 + 1 = 0$, we add 0011101010 and 1010110001 as follows.

$$\begin{array}{r} 0011101010 \\ + 1010110001 \\ \hline 1001011011 \end{array}$$

We refer to this operation as **word addition**. Word subtraction is the same as word addition. Exercise 17 shows that $\mathbb{B}^n$ is closed under word addition. A **binary group code** is any nonempty subset C of $\mathbb{B}^n$ that is closed under word addition. It can be shown that C has precisely 2^k elements for some integer k where $0 \leq k \leq n$. We refer to such a code as an (n, k) binary group code.

Encoding to Enable Correcting a Single-Error Transmission

We now show how, using more than one parity-check equation, we can not only detect but actually correct a single-error transmission of a code word. This method of encoding was developed by Richard Hamming in 1948.

Suppose we wish to encode the 16 message words

$$\begin{array}{cccccccc} 0000 & 0001 & 0010 & 0011 & 0100 & 0101 & 0110 & 0111 \\ 1000 & 1001 & 1010 & 1011 & 1100 & 1101 & 1110 & 1111 \end{array}$$

in $\mathbb{B}^4$ so that any single-error transmission of a code word not only can be detected but also corrected. The basic idea is simple. We append to the message word $x_1x_2x_3x_4$ some additional binary characters given by parity-check equations such as the equation $x_5 = x_1 + x_2 + x_3 + x_4$ in Illustration 2, and try to design the equations so that the minimum distance between the 16 code words created will be at least 3. Now, with a single-error transmission of a code word, the distance from the received word to that code word will be 1. If we can make our code words all at least three units apart, the pretransmission code word will be the unique code word at distance 1 from the received word. (If there were two such code words, the distance between them would be at most 2.)

In order to detect the error in a single-error transmission of a code word including not only message word characters but also the redundant parity-check characters, we need to have each component x_i in the code word appear at least once in *some* parity-check equation. Note that in Illustration 1, each component x_1, x_2, x_3, x_4 , and x_5 appears in the parity-check equation $x_5 = x_1 + x_2 + x_3 + x_4$. The parity-check equations

$$x_5 = x_1 + x_2 + x_3, \quad x_6 = x_1 + x_3 + x_4, \quad \text{and} \quad x_7 = x_2 + x_3 + x_4, \quad (2)$$

which we will show accomplish our goal, also satisfy this condition.

Let us see how to get a distance of at least 3 between each pair of the 16 code words. Of course, the distance between any two of the original 16 four-character message words is at least 1 because they are all different. Suppose now that two message words u and v differ in just one component, say x_i . A single parity-check equation containing x_i then yields a different character for u than for v . This shows that if each x_i in our original message word appears in at least two parity-check equations, then any message words at a distance of 1 are encoded into code words of distance at least 3. Note that the three parity-check equations [Eqs. (2)] satisfy this condition. It remains to ensure that two message words at a distance of 2 are encoded to increase this distance by at least 1. Suppose two message words u and v differ in only the i th and j th components. Now a parity-check equation containing both x_i and x_j will create the same parity-check character for u as for v . Thus, for each such combination i, j of positions in our message word, we need some parity-check equation to contain either x_i but not x_j or x_j but not x_i . We see that this condition is satisfied by the three parity-check equations [Eqs. (2)] for all possible combinations i, j —namely,

$$1,2 \quad 1,3 \quad 1,4 \quad 2,3 \quad 2,4 \quad \text{and} \quad 3,4.$$

Thus, these equations accomplish our goal. The 16 binary words of length 7, obtained by encoding the 16 binary words

$$\begin{array}{cccccccc} 0000 & 0001 & 0010 & 0011 & 0100 & 0101 & 0110 & 0111 \\ 1000 & 1001 & 1010 & 1011 & 1100 & 1101 & 1110 & 1111 \end{array}$$

of length 4 using Eqs. (2) form a subset of $\mathbb{B}^7$ called the Hamming (7, 4) code. In Exercise 16, we ask you to show that the Hamming (7, 4) code is a binary group code.

We can encode each of the 16 binary words of length 4 to form the Hamming (7, 4) code by multiplying the vector form $[x_1, x_2, x_3, x_4]$ of the word on the right by a 4×7 matrix G , called the **standard generator matrix**—namely, we compute

$$[x_1, x_2, x_3, x_4] \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \cdot G$$

To see this, note that the first four columns of G give the 4×4 identity matrix I , so the first four entries in the encoded word will yield precisely the message word $x_1x_2x_3x_4$. In columns 5, 6, and 7, we put the coefficients of x_1 , x_2 , x_3 , and x_4 as they appear in the parity-check equations defining x_5 , x_6 , and x_7 , respectively. Table 1.1 shows the 16 message words and the code words obtained using this generator matrix G . Note that the message words 0011 and 0111, which are at distance 1, have been encoded as 0011100 and 0111001, which are at distance 3. Also, the message words 0101 and 0011, which are at distance 2, have been encoded as 0101110 and 0011100, which are at distance 3.

Decoding a received word w by selecting a code word at minimum distance from w (in some codes, more than one code word might be at the minimum distance) is known as **nearest-neighbor decoding**. If transmission errors are independent from each other, it can be shown that this is equivalent to what is known as *maximum-likelihood decoding*.

ILLUSTRATION 4 Suppose the Hamming (7, 4) code shown in Table 1.1 is used. If the received word is 1011010, then the decoded message word consists of the first four characters 1011, because 1011010 is a code word. However, suppose the word 0110101 is received. This is not a code word. The closest code word is 0100101, which is at distance 1 from 0110101. Thus we decode 0110101 as

TABLE 1.1
The Hamming (7, 4) code

Message	Code Word
0000	0000000
0001	0001011
0010	0010111
0011	0011100
0100	0100101
0101	0101110
0110	0110010
0111	0111001
1000	1000110
1001	1001101
1010	1010001
1011	1011010
1100	1100011
1101	1101000
1110	1110100
1111	1111111

HISTORICAL NOTE RICHARD HAMMING (*b.* 1915) had his interest in the question of coding stimulated in 1947 when he was using an early Bell System relay computer on weekends only (because he did not have priority use of the machine). During the week, the machine sounded an alarm when it discovered an error so that an operator could attempt to correct it. On weekends, however, the machine was unattended and would dump any problem in which it discovered an error and proceed to the next one. Hamming's frustration with this behavior of the machine grew when errors cost him two consecutive weekends of work. He decided that if the machine could discover errors—it used a fairly simple error-detecting code—there must be a way for it to correct them and proceed with the solution. He therefore worked on this idea for the next year and discovered several different methods of creating error-correcting codes. Because of patent considerations, Hamming did not publish his solutions until 1950. A brief description of his (7, 4) code, however, appeared in a paper of Claude Shannon (*b.* 1916) in 1948.

Hamming, in fact, developed some of the parity-check ideas discussed in the text as well as the geometric model in which the distance between code words is the number of coordinates in which they differ. He also, in essence, realized that the set of actual code words embedded in $\mathbb{E}^7$ was a four-dimensional subspace of that space.

0100, which differs from the first four characters of the received word. On the other hand, if we receive the noncode word 1100111, we decode it as 1100, because the closest code word to 1100111 is 1100011. ■

Note in Illustration 4 that if the code word 0001011 is transmitted and is received as 0011001, with two errors, then the recipient knows that an error has been made, because 0011001 is not a code word. However, nearest-neighbor decoding yields the code word 0111001, which corresponds to a message word 0111 rather than the intended 0001. When retransmission is practical, it may be better to request it when an error is detected rather than blindly using nearest-neighbor decoding.

Of course, if errors are generated independently and transmission is of high quality, it should be much less likely for a word to be transmitted with two errors than with one error. If the probability of an error in transmission of a single character is p and errors are generated independently, then probability theory shows that in transmitting a word of length n ,

the probability of no error is $(1 - p)^n$,

the probability of exactly one error is $np(1 - p)^{n-1}$, and

the probability of exactly two errors is $\frac{n(n-1)}{2}p^2(1 - p)^{n-2}$.

For example, if $p = 0.0001$ so that we can expect about one character out of every 10,000 to be changed and if the length of the word is $n = 10$, then the probabilities of no error, one error, and two errors, respectively, are approximately 0.999, 0.000999, and 0.0000004.

Parity-Check Matrix Decoding

You can imagine that if we encoded all of the 256 ASCII characters in an $(n, 8)$ linear code and used nearest-neighbor decoding, it would be a job to pore over the list of 256 encoded characters to determine the nearest neighbor to a received code word. There is an easier way, which we illustrate using the Hamming (7, 4) code developed before Illustration 4. Recall that the parity-check equations for that code are

$$x_5 = x_1 + x_2 + x_3, \quad x_6 = x_1 + x_3 + x_4, \quad \text{and} \quad x_7 = x_2 + x_3 + x_4.$$

Let us again concern ourselves with detecting and correcting just single-error transmissions. If these parity-check equations hold for the received word, then no such single error has occurred. Suppose, on the other hand, that the first two equations fail and the last one holds. The only character appearing in both of the first two equations but not in the last is x_1 , so we could simply change the character x_1 from 0 to 1, or vice versa, to decode. Note that each of x_1 , x_2 , and x_4 is omitted just once but in different equations, x_1 is the only character that appears in all three equations, and each of x_5 , x_6 , and x_7 appears just once but in different equations. This allows us to identify the character in a single-error transmission easily.

We can be even more systematic. Let us rewrite the equations as

$$\begin{aligned}x_1 + x_2 + x_3 + x_5 &= 0, & x_1 + x_3 + x_4 + x_6 &= 0, & \text{and} \\x_2 + x_3 + x_4 + x_7 &= 0.\end{aligned}$$

We form the **parity-check matrix** H whose i th row contains the seven coefficients of $x_1, x_2, x_3, x_4, x_5, x_6, x_7$ in the i th equation—namely,

$$H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}.$$

Let w be a received word, written as a column vector. Exercise 26 shows that w is a code word if and only if Hw is the zero column vector, where we are always using modulo 2 arithmetic. If w resulted from a single-error transmission in which the character in the j th position was changed, then Hw would have 1 in its i th component if and only if x_j appeared in the i th parity-check equation, so that the column vector Hw would be the j th column of H . Thus we can decode a received word w as follows in the Hamming (7, 4) code of Illustration 4, and be confident of detecting and correcting any single-position error, as follows.

Parity-Check Matrix Decoding

1. Compute Hw .
2. If Hw is the zero vector, decode as the first four characters of w .
3. If Hw is the j th column of H , then:
 - a. if $j > 4$, decode as the first four characters of w ;
 - b. otherwise, decode as the first four characters with the j th character changed.
4. If Hw is neither the zero vector nor the j th column of H , then more than one error has been made: ask for retransmission.

ILLUSTRATION 5 Suppose the Hamming (7, 4) code shown in Table 1.1 on page 120 is used and the word $w = 0110101$ is received. We compute that

$$Hw = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Because this is the third column of H , we change the third character in the message portion 0110 and decode as 0100. Note that this is what we obtained in Illustration 4 when we decoded this word using Table 1.1. ■

Just as we did after Illustration 4, we point out that if two errors are made in transmission, the preceding outline may lead to incorrect decoding. If the code word $v = 0001011$ is transmitted and received as $w = 0011001$ with two errors, then

$$Hw = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},$$

which is column 2 of the matrix H . Thus, decoding by the steps above leads to the incorrect message word 0111. Note that item 4 in the list above does not say that if more than one error has been made, then Hw is neither the zero vector nor the j th column of H .

EXERCISES

- Let the binary numbers 1 through 15 stand for the letters $A B C D E F G H I J K L M N O$, in that order. Using Table 1.1 for the Hamming (7, 4) code and letting 0000 stand for a space between words, encode the message *A GOOD DOG*.
- With the same understanding as in the preceding exercise, use nearest-neighbor decoding to decode this received message.

```
0111001 1111111 1010100 0101110
0000000 1100110 1111111 1101000
1101110
```

In Exercises 3 through 11, consider the (6, 3) linear code C with standard generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

- Give the parity-check equations for this code.
- List the code words in C .
- How many errors can always be detected using this code?
- How many errors can always be corrected using this code?
- Assuming that the given word has been received, decode it using nearest-neighbor decoding, using your list of code words in

Exercise 4. (Recall that in the case where more than one code word is at minimum distance from the received word, a code word of minimum distance is selected arbitrarily.)

- 110111
- 001011
- 111011
- 101010
- 100101

- Give the parity-check matrix for this code.
- Use the parity-check matrix to decode the received words in Exercise 7.
- Let $u = 110101011$ and $v = 0111001110$. Find
 - $\text{wt}(u)$
 - $\text{wt}(v)$
 - $u + v$
 - $d(u, v)$
- Show that for word addition of binary words u and v of the same length, we have $u + v = u - v$.
- If a binary code word u is transmitted and the received word is w , then the sum $u + w$ given by word addition modulo 2 is called the *error pattern*. Explain why this is a descriptive name for this sum.
- Show that for two binary words of the same length, we have $d(u, v) = \text{wt}(u - v)$.

14. Prove the following properties of the distance function for binary words u , v , and w of the same length.
- $d(u, v) = 0$ if and only if $u = v$
 - $d(u, v) = d(v, u)$ (Symmetry)
 - $d(u, w) \leq d(u, v) + d(v, w)$ (Triangle inequality)
 - $d(u, v) = d(u + w, v + w)$ (Invariance under translation)
15. Show that $\mathbb{B}^n$ is closed under word addition.
16. Recall that we call a nonempty subset C of $\mathbb{B}^n$ a *binary group code* if C is closed under addition. Show that the Hamming (7, 4) code is a group code. [HINT: To show closure under word addition, use the fact that the words in the Hamming (7, 4) code can be formed from those in $\mathbb{B}^4$ by multiplying by a generator matrix.]
17. Show that in a binary group code C , the minimum distance between code words is equal to the minimum weight of the nonzero code words.
18. Suppose that you want to be able to recognize that a received word is incorrect when m or fewer of its characters have been changed during transmission. What must be the minimum distance between code words to accomplish this?
19. Suppose that you want to be able to find a unique nearest neighbor for a received word that has been transmitted with m or fewer of its characters changed. What must be the minimum distance between code words to accomplish this?
20. Show that if the minimum nonzero weight of code words in a group code C is at least $2t + 1$, then the code can detect any $2t$ errors and correct any t errors. (Compare the result stated in this exercise with your answers to the two preceding ones.)
21. Show that if the minimum distance between the words in an (n, k) binary group code C is at least 3, we must have

$$2^{n-k} \geq 1 + n.$$

[HINT: Let e_i be the word in $\mathbb{B}^n$ with 1 in the i th position and 0's elsewhere. Show that e_i is not in C and that, for any two distinct words v and w in C , we have $v + e_i \neq w + e_i$

if $i \neq j$. Then use the fact that n must be large enough so that $\mathbb{B}^n$ contains C and all words whose distance from some word in C is 1.]

22. Show that if the minimum distance between the words in an (n, k) binary group code C is at least 5, then we must have

$$2^{n-k} \geq 1 + n + \frac{n(n-1)}{2}.$$

[HINT: Proceed as suggested by the hint in Exercise 21 to count the words at distance 1 and at distance 2 from some word in C .]

23. Using the formulas in Exercises 21 and 22, find a lower bound for the number of parity-check equations necessary to encode the 2^k words in $\mathbb{B}^k$ so that the minimum distance between different code words is at least m for the given values of m and k . (Note that $k = 8$ would allow us to encode all the ASCII characters, and that $m = 5$ would allow us to detect and correct all single-error and double-error transmissions using nearest-neighbor decoding.)
- $k = 2, m = 3$
 - $k = 4, m = 3$
 - $k = 8, m = 3$
 - $k = 2, m = 5$
 - $k = 4, m = 5$
 - $k = 8, m = 5$
24. Find parity-check equations for encoding the 32 words in $\mathbb{B}^5$ into an $(n, 5)$ linear code that can be used to detect and correct any single-error transmission of a code word. (Recall that each character x_i must appear in two parity-check equations, and that for each pair x_i, x_j some equation must contain one of them but not the other.) Try to make the number of parity-check equations as small as possible; see Exercise 21. Give the standard generator matrix for your code.
25. The 256 ASCII characters are numbered from 0 to 255, and thus can be represented by the 256 binary words in $\mathbb{B}^8$. Find $n - 8$ parity-check equations that can be used to form an $(n, 8)$ linear code that can be used to detect and correct any single-error transmission of a code word. Try to make n the value found in part (c) of Exercise 23.
26. Let C be an (n, k) linear code with parity-check matrix H . We know that $Hc = 0$ for all $c \in C$. Show conversely that if $w \in \mathbb{B}^n$ and $Hw = 0$, then $w \in C$.

DIMENSION, RANK, AND LINEAR TRANSFORMATIONS

Given a finite set S of vectors that generate a subspace W of $\mathbb{R}^n$, we would like to delete from S any superfluous vectors, obtaining as small a subset B of S as we can that still generates W . We tackle this problem in Section 2.1. In doing so, we encounter the notion of an *independent set* of vectors. We discover that such a minimal subset B of S that generates W is a basis for W , so that every vector in W can be expressed uniquely as a linear combination of vectors in B . We will see that any two bases for W contain the same number of vectors—the *dimension* of W will be defined to be this number. Section 2.2 discusses the relationships among the dimensions of the column space, the row space, and the nullspace of a matrix.

In Section 2.3, we discuss functions mapping $\mathbb{R}^n$ into $\mathbb{R}^m$ that *preserve*, in a sense that we will describe, both vector addition and scalar multiplication. Such functions are known as *linear transformations*. We will see that for a linear transformation, the image of a vector x in $\mathbb{R}^n$ can be computed by multiplying the column vector x on the left by a suitable $m \times n$ matrix. Optional Section 2.4 then applies matrix techniques in describing geometrically all linear transformations of the plane $\mathbb{R}^2$ into itself.

As another application to geometry, optional Section 2.5 uses vector techniques to generalize the notions of line and plane to k -dimensional flats in $\mathbb{R}^n$.

2.1

INDEPENDENCE AND DIMENSION

Finding a Basis for a Span of Vectors

Let $w_1, w_2, \dots, w_k$ be vectors in $\mathbb{R}^n$ and let $W = \text{sp}(w_1, w_2, \dots, w_k)$. Now W can be characterized as the *smallest* subspace of $\mathbb{R}^n$ containing all of the vectors $w_1, w_2, \dots, w_k$, because every subspace containing these vectors must contain all

linear combinations of them, and consequently must include every vector in W . We set ourselves the problem of finding a basis for W .

Let us assume that $\{w_1, w_2, \dots, w_k\}$ is not a basis for W . Theorem 1.15 then tells us that we can express the zero vector as a linear combination of the w_j in some nontrivial way. As an illustration, suppose that

$$2w_2 - 5w_6 + \frac{1}{3}w_7 = 0. \quad (1)$$

Using Eq. (1), we can express each of w_2 , w_6 , and w_7 as a linear combination of the other two. For example, we have

$$\frac{1}{3}w_7 = -2w_2 + 5w_6 \quad \text{so} \quad w_7 = -6w_2 + 15w_6.$$

We claim that we can delete w_7 from our list $w_1, w_2, \dots, w_k$, and the remaining w_j will still span W . The space spanned by the remaining w_j , which is contained in W , will still contain w_7 because $w_7 = -6w_2 + 15w_6$, and we have seen that W is the *smallest space* containing all the w_j . Thus the vector w_7 in the original list is not needed to span W .

The preceding illustration indicates that we can find a basis for $W = \text{sp}(w_1, w_2, \dots, w_k)$ by repeatedly deleting from the list of $w_1, w_2, \dots, w_k$ one vector that appears with a nonzero coefficient in a linear combination giving the zero vector, such as Eq. (1), until no such nontrivial linear combination for 0 exists. The final list of remaining w_j will still span W and be a basis for W by Theorem 1.15.

EXAMPLE 1 Find a basis for $W = \text{sp}([2, 3], [0, 1], [4, -6])$ in $\mathbb{R}^2$.

SOLUTION The presence of the vector $[0, 1]$ allows us to spot that $[4, -6] = 2[2, 3] - 12[0, 1]$, so we have a relation like Eq. (1)—namely,

$$2[2, 3] - 12[0, 1] - [4, -6] = [0, 0].$$

Thus we can delete any one of the three vectors, and the remaining two will still span W . For example, we can delete the vector $[4, -6]$ and we will have $W = \text{sp}([2, 3], [0, 1])$. Because neither of these two remaining vectors is a multiple of the other, we see that the zero vector cannot be expressed as a nontrivial linear combination of them. (See Exercise 29.) Thus $\{[2, 3], [0, 1]\}$ is a basis for W , which we realize is actually all of $\mathbb{R}^2$, because any two nonzero and nonparallel vectors span $\mathbb{R}^2$ (Theorem 1.16 in Section 1.6). ■

Our attention is focused on the existence of a nontrivial linear combination yielding the zero vector, such as Eq. (1). Such an equation is known as a *dependence relation*. We formally define this, and the notions of *dependence* and *independence* for vectors. This is a very important definition in our study of linear algebra.

DEFINITION 2.1 Linear Dependence and Independence

Let $\{w_1, w_2, \dots, w_k\}$ be a set of vectors in $\mathbb{R}^n$. A **dependence relation** in this set is an equation of the form

$$r_1 w_1 + r_2 w_2 + \dots + r_k w_k = 0, \quad \text{with at least one } r_j \neq 0. \quad (2)$$

If such a dependence relation exists, then $\{w_1, w_2, \dots, w_k\}$ is a **linearly dependent** set of vectors. Otherwise, the set of vectors is **linearly independent**.

For convenience, we will often drop the word *linearly* from the terms *linearly dependent* and *linearly independent*, and just speak of a *dependent* or *independent set of vectors*. We will sometimes drop the words *set of* and refer to *dependent* or *independent vectors* $w_1, w_2, \dots, w_k$.

Two nonzero vectors in $\mathbb{R}^n$ are independent if and only if one is not a scalar multiple of the other (see Exercise 29). Figure 2.1(a) shows two independent vectors w_1 and w_2 in $\mathbb{R}^n$. A little thought shows why $r_1 w_1 + r_2 w_2$ in this figure can be the zero vector if and only if $r_1 = r_2 = 0$. Figure 2.1(b) shows three independent vectors w_1, w_2 , and w_3 in $\mathbb{R}^n$. Note how $w_1 \notin \text{sp}(w_2, w_3)$. Similarly, $w_2 \notin \text{sp}(w_1, w_3)$ and $w_3 \notin \text{sp}(w_1, w_2)$.

Using our new terminology, Theorem 1.15 shows that $\{w_1, w_2, \dots, w_k\}$ is a basis for a subspace W of $\mathbb{R}^n$ if and only if the vectors $w_1, w_2, \dots, w_k$ span W and are independent. This is taken as a definition of a basis in many texts. We chose the “unique linear combination” characterization in Definition 1.17 because it is the most important property of bases and was the natural choice arising from our discussion of the solution set of $Ax = 0$. We state this alternative characterization as a theorem.

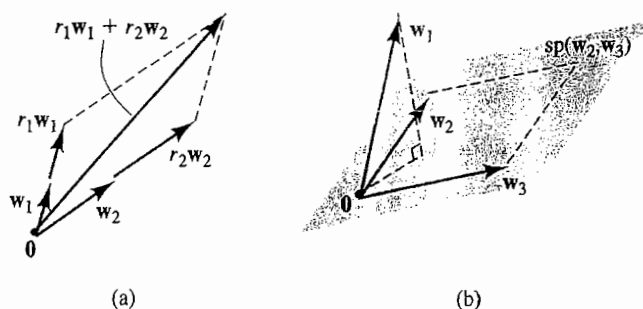


FIGURE 2.1
(a) Independent vectors w_1 and w_2 ; (b) independent vectors w_1, w_2 , and w_3 .

THEOREM 2.1 Alternative Characterization of a Basis

Let W be a subspace of $\mathbb{R}^n$. A subset $\{w_1, w_2, \dots, w_k\}$ of W is a basis for W if and only if the following two conditions are met:

1. The vectors $w_1, w_2, \dots, w_k$ span W .
2. The vectors are linearly independent.

We turn to a technique for computing a basis for $W = \text{sp}(w_1, w_2, \dots, w_k)$ in $\mathbb{R}^n$. Determining whether there is a nontrivial dependence relation

$$x_1 w_1 + x_2 w_2 + \dots + x_k w_k = 0, \quad \text{some } x_j \neq 0,$$

amounts to determining whether the linear system $Ax = 0$ has a nontrivial solution, where A is the matrix whose j th column vector is w_j . The ubiquitous row reduction appears again! This time, we will get much more information than just the existence of a dependence relation. Recall that the solutions of $Ax = 0$ are identical with those of the system $Hx = 0$ where $[H | 0]$ is obtained from $[A | 0]$ by row reduction. Suppose, for illustration, that H is in reduced row-echelon form, and that

$$[H | 0] = \left[\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & 5 & 0 \\ 0 & 1 & -3 & 0 & 9 & 0 \\ 0 & 0 & 0 & 1 & -7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

(Normally, we would not bother to write the column of zeros to the right of the partition, but we want to be sure that you realize that, in this context, we imagine the zero column to be there.) The zeros above as well as below the pivots allow us to spot some dependence relations (arising from solutions of $Hx = 0$) immediately, because the columns with pivots are in standard basis vector form. Every nontrivial solution of $Hx = 0$ is a nontrivial solution of $Ax = 0$ and so gives a dependence relation on the column vectors w_j of A . In particular, we see that

$$w_3 = 2w_1 - 3w_2 \quad \text{and} \quad w_5 = 5w_1 + 9w_2 - 7w_4,$$

and so we have the dependence relations

$$-2w_1 + 3w_2 + w_3 = 0 \quad \text{and} \quad -5w_1 - 9w_2 + 7w_4 + w_5 = 0.$$

Hence we can delete w_3 and w_5 and retain $\{w_1, w_2, w_4\}$ as a basis for W . In order to be systematic, we have chosen to keep precisely the vectors w_j such that the j th column of H contains a pivot. Thus we don't really have to obtain *reduced* row-echelon form with zeros above pivots to do this; row-echelon form is enough. We have hit upon the following elegant technique.

Finding a Basis for $W = \text{sp}(w_1, w_2, \dots, w_k)$

1. Form the matrix A whose j th column vector is w_j .
2. Row-reduce A to row-echelon form H .
3. The set of all w_j such that the j th column of H contains a pivot is a basis for W .

EXAMPLE 2 Find a basis for the subspace W of $\mathbb{R}^5$ spanned by

$$\begin{aligned} w_1 &= [1, -1, 0, 2, 1], & w_2 &= [2, 1, -2, 0, 0], \\ w_3 &= [0, -3, 2, 4, 2], & w_4 &= [3, 3, -4, -2, -1], \\ w_5 &= [2, 4, 1, 0, 1], & w_6 &= [5, 7, -3, -2, 0]. \end{aligned}$$

SOLUTION We reduce the matrix that has w_j as j th column vector, obtaining

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 0 & 3 & 2 & 5 \\ -1 & 1 & -3 & 3 & 4 & 7 \\ 0 & -2 & 2 & -4 & 1 & -3 \\ 2 & 0 & 4 & -2 & 0 & -2 \\ 1 & 0 & 2 & -1 & 1 & 0 \end{bmatrix} &\sim \begin{bmatrix} 1 & 2 & 0 & 3 & 2 & 5 \\ 0 & 3 & -3 & 6 & 6 & 12 \\ 0 & -2 & 2 & -4 & 1 & -3 \\ 0 & -4 & 4 & -8 & -4 & -12 \\ 0 & -2 & 2 & -4 & -1 & -5 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 2 & 0 & 3 & 2 & 5 \\ 0 & 1 & -1 & 2 & 2 & 4 \\ 0 & 0 & 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 0 & 3 & 3 \end{bmatrix} &\sim \begin{bmatrix} 1 & 2 & 0 & 3 & 2 & 5 \\ 0 & 1 & -1 & 2 & 2 & 4 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

Because there are pivots in columns 1, 2, and 5 of the row-echelon form, the vectors w_1 , w_2 , and w_5 are retained and are independent. We obtain $\{w_1, w_2, w_5\}$ as a basis for W . ■

We emphasize that the vectors retained are in columns of the matrix A formed at the start of the reduction process. A common error is to take instead the actual column vectors containing pivots in the row-echelon form H . There is no reason even to expect that the column vectors of H lie in the column space of A . For example,

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = H,$$

and certainly $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is not in the column space of A .

Note that we can test whether vectors in $\mathbb{R}^n$ are independent by reducing a matrix having them as column vectors. The vectors are independent if and only if row reduction of the matrix yields a matrix with a pivot in every column. In particular, n vectors in $\mathbb{R}^n$ are independent if and only if row

reduction of the matrix having them as column vectors yields the $n \times n$ identity matrix I . On the other hand, more than n vectors in $\mathbb{R}^n$ must be dependent, because an $m \times n$ matrix with $m < n$ cannot have a pivot in every column.

EXAMPLE 3 Determine whether the vectors $\mathbf{v}_1 = [1, 2, 3, 1]$, $\mathbf{v}_2 = [2, 2, 1, 3]$, and $\mathbf{v}_3 = [-1, 2, 7, -3]$ in $\mathbb{R}^4$ are independent.

SOLUTION Reducing the matrix with j th column vector $\mathbf{v}_j$, we obtain

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 2 & 2 \\ 3 & 1 & 7 \\ 1 & 3 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 4 \\ 0 & -5 & 10 \\ 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

We see that the vectors are not independent. In fact, $\mathbf{v}_3 = 3\mathbf{v}_1 - 2\mathbf{v}_2$. ■

The Dimension of a Subspace

We realize that a basis for a subspace W of $\mathbb{R}^n$ is far from unique. In Example 1, for $W = \text{sp}([2, 3], [0, 1], [4, -6])$ in $\mathbb{R}^2$, we discovered that any two of the three vectors can be used to form a basis for W . We also know that any two nonzero, nonparallel vectors in $\mathbb{R}^2$ form a basis for $\mathbb{R}^2$. Likewise, in Example 2, the vectors we found for a basis for $\text{sp}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4, \mathbf{w}_5, \mathbf{w}_6)$ depended on the order in which we put them as columns in the matrix A that we row-reduced. If we reverse the order of the columns in the matrix A in Example 2, we will wind up with a different basis. However, it is true that given a subspace W of $\mathbb{R}^n$, all bases for W contain the same number of vectors. This is an important result, which will be a quick corollary of the following theorem.

THEOREM 2.2 Relative Sizes of Spanning and Independent Sets

Let W be a subspace of $\mathbb{R}^n$. Let $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k$ be vectors in W that span W , and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ be vectors in W that are independent. Then $k \geq m$.

PROOF Let us suppose that $k < m$. We will show that the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ are dependent, contrary to hypothesis. Because the vectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k$ span W , there exist scalars a_{ij} such that

$$\begin{aligned} \mathbf{v}_1 &= a_{11}\mathbf{w}_1 + a_{21}\mathbf{w}_2 + \cdots + a_{k1}\mathbf{w}_k \\ \mathbf{v}_2 &= a_{12}\mathbf{w}_1 + a_{22}\mathbf{w}_2 + \cdots + a_{k2}\mathbf{w}_k \\ &\vdots \\ \mathbf{v}_m &= a_{1m}\mathbf{w}_1 + a_{2m}\mathbf{w}_2 + \cdots + a_{km}\mathbf{w}_k \end{aligned} \tag{3}$$

We compute $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_m\mathbf{v}_m$ in an attempt to find a dependence relation by multiplying the first equation in Eqs. (3) by x_1 , the second by x_2 ,

etc., and then adding the equations. Now the resulting sum is sure to be the zero vector if the total coefficient of each w_j on the right-hand side in the sum after adding is zero—that is, if we can make

$$x_1 a_{11} + x_2 a_{12} + \cdots + x_m a_{1m} = 0,$$

$$x_1 a_{21} + x_2 a_{22} + \cdots + x_m a_{2m} = 0,$$

$$\vdots$$

$$x_1 a_{k1} + x_2 a_{k2} + \cdots + x_m a_{km} = 0.$$

This gives us a homogeneous linear system of k equations in m unknowns x_i . Corollary 2 of Theorem 1.17 tells us that such a homogeneous system has a nontrivial solution if there are fewer equations than unknowns, and this is the case because we are supposing that $k < m$. Thus we can find scalars $x_1, x_2, \dots, x_m$, not all zero, such that $x_1 v_1 + x_2 v_2 + \cdots + x_m v_m = \mathbf{0}$. That is, the vectors $v_1, v_2, \dots, v_m$ are dependent if $k < m$, as we wanted to show. $\blacktriangle$

COROLLARY Invariance of Dimension

Any two bases of a subspace W of $\mathbb{R}^n$ contain the same number of vectors.

PROOF Suppose that both a set B with k vectors and a set B' with m vectors are bases for W . Then both B and B' are independent sets of vectors, and the vectors in either set span W . Regarding B as a set of k vectors spanning W and regarding B' as a set of m independent vectors in W , Theorem 2.2 tells us that $k \geq m$. Switching around and regarding B' as a set of m vectors spanning W and regarding B as a set of k independent vectors in W , the theorem tells us that $m \geq k$. Therefore, $k = m$. $\blacktriangle$

As the title of the corollary indicates, we will consider the number of vectors in a basis for a subspace W of $\mathbb{R}^n$ to be the *dimension* of W . If different bases for W were to have different numbers of vectors, then this notion of dimension would not be well defined (that is, unambiguous). Because different people may come up with different bases, the preceding corollary is necessary in order to define *dimension*.

DEFINITION 2.2 Dimension of a Subspace

Let W be a subspace of $\mathbb{R}^n$. The number of elements in a basis for W is the **dimension** of W , and is denoted by $\dim(W)$.

Thus the dimension of $\mathbb{R}^n$ is n , because we have the standard basis $\{e_1, e_2, \dots, e_n\}$. Now $\mathbb{R}^n$ cannot be spanned by fewer than n vectors, because a

spanning set can always be cut down (if necessary) to form a basis using the technique boxed before Example 2. Theorem 2.2 also tells us that we cannot find a set containing more than n independent vectors in $\mathbb{R}^n$. The same observations hold for any subspace W of $\mathbb{R}^n$ using the same arguments. If $\dim(W) = k$, then W cannot be spanned by fewer than k vectors, and an independent set of vectors in W can contain at most k elements. Perhaps you just assumed that this would be the case; it is gratifying now to have justification for it.

EXAMPLE 4 Find the dimension of the subspace $W = \text{sp}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4)$ of $\mathbb{R}^3$ where $\mathbf{w}_1 = [1, -3, 1]$, $\mathbf{w}_2 = [-2, 6, -2]$, $\mathbf{w}_3 = [2, 1, -4]$, and $\mathbf{w}_4 = [-1, 10, -7]$.

SOLUTION Clearly, $\dim(W)$ is no larger than 3. To determine its value, we form the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 & -1 \\ -3 & 6 & 1 & 10 \\ 1 & -2 & -4 & -7 \end{bmatrix}.$$

We reduce the matrix A to row-echelon form, obtaining

$$\begin{bmatrix} 1 & -2 & 2 & -1 \\ -3 & 6 & 1 & 10 \\ 1 & -2 & -4 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 2 & -1 \\ 0 & 0 & 7 & 7 \\ 0 & 0 & -6 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Thus the column vectors

$$\begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix}$$

form a basis for W , the column space of A , and so $\dim(W) = 2$. ■

In Section 1.6, we stated that we would show that every subspace W of $\mathbb{R}^n$ is of the form $\text{sp}(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k)$. We do this now by showing that every subspace $W \neq \{0\}$ has a basis. Of course, $\{0\} = \text{sp}(0)$. To construct a basis for $W \neq \{0\}$, choose any nonzero vector $\mathbf{w}_1$ in W . If $W = \text{sp}(\mathbf{w}_1)$, we are done. If not, choose a vector $\mathbf{w}_2$ in W that is not in $\text{sp}(\mathbf{w}_1)$. Now the vectors $\mathbf{w}_1, \mathbf{w}_2$ must be independent, for a dependence relation would allow us to express $\mathbf{w}_2$ as a multiple of $\mathbf{w}_1$, contrary to our choice of $\mathbf{w}_2$ not in $\text{sp}(\mathbf{w}_1)$. If $\text{sp}(\mathbf{w}_1, \mathbf{w}_2) = W$, we are done. If not, choose $\mathbf{w}_3 \in W$ that is not in $\text{sp}(\mathbf{w}_1, \mathbf{w}_2)$. Again, no dependence relation can exist for $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ because none exists for $\mathbf{w}_1, \mathbf{w}_2$ and because $\mathbf{w}_3$ cannot be a linear combination of $\mathbf{w}_1$ and $\mathbf{w}_2$. Continue in this fashion. Now W cannot contain an independent set with more than n vectors because no independent subset of $\mathbb{R}^n$ can have more than n vectors (Theorem 2.2). The process must stop with $W = \text{sp}(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k)$ for some $k \leq n$, which demonstrates our goal. In order to be able to say that every subspace of $\mathbb{R}^n$ has a basis, we *define* the basis of the zero subspace $\{0\}$ to be the empty set. Note that although $\text{sp}(0) = \{0\}$, the zero vector is not a *unique* linear com-

bination of itself, because $r\mathbf{0} = \mathbf{0}$ for all scalars r . In view of our definition, we have $\dim(\{\mathbf{0}\}) = 0$.

The construction technique in the preceding paragraph also shows that every independent subset S of vectors in a subspace W of $\mathbb{R}^n$ can be enlarged, if necessary, to become a basis for W . Namely, if S is not already a basis, we choose a vector in W that is not in the span of the vectors in S , enlarge S by this vector, and continue this process until S becomes a basis.

If we know already that $\dim(W) = k$ and want to check that a subset S containing k vectors of W is a basis, it is not necessary to check both (1) that the vectors in S span W and (2) that they are independent. It suffices to check just one of these conditions. Because if the vectors span S , we know that the set S can be cut down—if necessary, by the technique of Example 2—to become a basis. Because S already has the required number of vectors for a basis, no such cutting down can occur. On the other hand, if we know that S is an independent set, then the preceding paragraph shows that S can be enlarged, if necessary, to become a basis. But because S has the right number of vectors for a basis, no such enlargement is possible.

We collect the observations in the preceding three paragraphs in a theorem for easy reference.

THEOREM 2.3 Existence and Determination of Bases

1. Every subspace W of $\mathbb{R}^n$ has a basis and $\dim(W) \leq n$.
2. Every independent set of vectors in $\mathbb{R}^n$ can be enlarged, if necessary, to become a basis for $\mathbb{R}^n$.
3. If W is a subspace of $\mathbb{R}^n$ and $\dim(W) = k$, then
 - a. every independent set of k vectors in W is a basis for W , and
 - b. every set of k vectors in W that spans W is a basis for W .

The example that follows illustrates a technique for enlarging an independent set of vectors in $\mathbb{R}^n$ to a basis for $\mathbb{R}^n$.

EXAMPLE 5 Enlarge the independent set $\{[1, 1, -1], [1, 2, -2]\}$ to a basis for $\mathbb{R}^3$.

SOLUTION Let $\mathbf{v}_1 = [1, 1, -1]$ and $\mathbf{v}_2 = [1, 2, -2]$. We know a spanning set for $\mathbb{R}^3$ —namely, $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$. We write $\mathbb{R}^3 = \text{sp}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ and apply the technique of Example 2 to find a basis. As long as we put $\mathbf{v}_1$ and $\mathbf{v}_2$ first as columns of the matrix to be reduced, pivots will occur in those columns, so $\mathbf{v}_1$ and $\mathbf{v}_2$ will be retained in the basis. We obtain

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 \\ -1 & -2 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & -1 & 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

We see that the pivots occur in columns 1, 2, and 4. Thus a basis containing $\mathbf{v}_1$ and $\mathbf{v}_2$ is $\{[1, 1, -1], [1, 2, -2], [0, 1, 0]\}$. ■

SUMMARY

1. A set of vectors $\{w_1, w_2, \dots, w_k\}$ in $\mathbb{R}^n$ is *linearly dependent* if there exists a *dependence relation*

$$r_1 w_1 + r_2 w_2 + \dots + r_k w_k = 0, \quad \text{with at least one } r_j \neq 0.$$

The set is *linearly independent* if no such dependence relation exists, so that a linear combination of the w_i is the zero vector only if all of the scalar coefficients are zero.

2. A set B of vectors in a subspace W of $\mathbb{R}^n$ is a *basis* for W if and only if the set is independent and the vectors span W . Equivalently, each vector in W can be written *uniquely* as a linear combination of the vectors in B .
3. If $W = \text{sp}(w_1, w_2, \dots, w_k)$, then the set $\{w_1, w_2, \dots, w_k\}$ can be cut down, if necessary, to a basis for W by reducing the matrix A having w_j as the j th column vector to row-echelon form H , and retaining w_j if and only if the j th column of H contains a pivot.
4. Every subspace W of $\mathbb{R}^n$ has a basis, and every independent set of vectors in W can be enlarged (if necessary) to a basis for W .
5. Let W be a subspace of $\mathbb{R}^n$. All bases of W contain the same number of vectors. The *dimension* of W , denoted by $\dim(W)$, is the number of vectors in any basis for W .
6. Let W be a subspace of $\mathbb{R}^n$ and let $\dim(W) = k$. A subset S of W containing exactly k vectors is a basis for W if either
 - a. S is an independent set, or
 - b. S spans W .

That is, it is not necessary to check both conditions in Theorem 2.1 for a basis if S has the right number of elements for a basis.

EXERCISES

1. Give a geometric criterion for a set of two distinct nonzero vectors in $\mathbb{R}^2$ to be dependent.
6. Argue geometrically that every set of four distinct vectors in $\mathbb{R}^3$ is dependent.

2. Argue geometrically that any set of three distinct vectors in $\mathbb{R}^3$ is dependent.
3. Give a geometric criterion for a set of two distinct nonzero vectors in $\mathbb{R}^3$ to be dependent.
4. Give a geometric description of the subspace of $\mathbb{R}^3$ generated by an independent set of two vectors.
5. Give a geometric criterion for a set of three distinct nonzero vectors in $\mathbb{R}^3$ to be dependent.

In Exercises 7–11, use the technique of Example 2, described in the box on page 129, to find a basis for the subspace spanned by the given vectors.

7. $\text{sp}([-3, 1], [6, 4])$ in $\mathbb{R}^2$
8. $\text{sp}([-3, 1], [9, -3])$ in $\mathbb{R}^2$
9. $\text{sp}([2, 1], [-6, -3], [1, 4])$ in $\mathbb{R}^2$
10. $\text{sp}([-2, 3, 1], [3, -1, 2], [1, 2, 3], [-1, 5, 4])$ in $\mathbb{R}^3$

11. $\text{sp}([1, 2, 1, 2], [2, 1, 0, -1], [-1, 4, 3, 8], [0, 3, 2, 5])$ in $\mathbb{R}^4$
12. Find a basis for the column space of the matrix

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 2 & 1 \\ 1 & 7 & 2 \\ 6 & -2 & 0 \end{bmatrix}$$

13. Find a basis for the row space of the matrix

$$A = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 0 & 4 & 2 \\ 3 & 2 & 8 & 7 \end{bmatrix}$$

14. Find a basis for the column space of the matrix A in Exercise 13.
15. Find a basis for the row space of the matrix A in Exercise 12.

In Exercises 16–25, use the technique illustrated in Example 3 to determine whether the given set of vectors is dependent or independent.

16. $\{[1, 3], [-2, -6]\}$ in $\mathbb{R}^2$
17. $\{[1, 3], [2, -4]\}$ in $\mathbb{R}^2$
18. $\{[-3, 1], [6, 4]\}$ in $\mathbb{R}^2$
19. $\{[-3, 1], [9, -3]\}$ in $\mathbb{R}^2$
20. $\{[2, 1], [-6, -3], [1, 4]\}$ in $\mathbb{R}^3$
21. $\{[-1, 2, 1], [2, -4, 3]\}$ in $\mathbb{R}^3$
22. $\{[1, -3, 2], [2, -5, 3], [4, 0, 1]\}$ in $\mathbb{R}^3$
23. $\{[1, -4, 3], [3, -11, 2], [1, -3, -4]\}$ in $\mathbb{R}^3$
24. $\{[1, 4, -1, 3], [-1, 5, 6, 2], [1, 13, 4, 7]\}$ in $\mathbb{R}^4$
25. $\{[-2, 3, 1], [3, -1, 2], [1, 2, 3], [-1, 5, 4]\}$ in $\mathbb{R}^3$

In Exercises 26 and 27, enlarge the given independent set to a basis for the entire space $\mathbb{R}^n$.

26. $\{[1, 2, 1]\}$ in $\mathbb{R}^3$
27. $\{[2, 1, 1, 1], [1, 0, 1, 1]\}$ in $\mathbb{R}^4$
28. Let $S = \{v_1, v_2, \dots, v_k\}$ be a set of vectors in $\mathbb{R}^n$. Mark each of the following True or False.
- a. A subset of $\mathbb{R}^n$ containing two nonzero distinct parallel vectors is dependent.

- b. If a set of nonzero vectors in $\mathbb{R}^n$ is dependent, then any two vectors in the set are parallel.
- c. Every subset of three vectors in $\mathbb{R}^2$ is dependent.
- d. Every subset of two vectors in $\mathbb{R}^2$ is independent.
- e. If a subset of two vectors in $\mathbb{R}^2$ spans $\mathbb{R}^2$, then the subset is independent.
- f. Every subset of $\mathbb{R}^n$ containing the zero vector is dependent.
- g. If S is independent, then each vector in $\mathbb{R}^n$ can be expressed uniquely as a linear combination of vectors in S .
- h. If S is independent and spans $\mathbb{R}^n$, then each vector in $\mathbb{R}^n$ can be expressed uniquely as a linear combination of vectors in S .
- i. If each vector in $\mathbb{R}^n$ can be expressed uniquely as a linear combination of vectors in S , then S is an independent set.
- j. The subset S is independent if and only if each vector in $\text{sp}(v_1, v_2, \dots, v_k)$ has a unique expression as a linear combination of vectors in S .
- k. The zero subspace of $\mathbb{R}^n$ has dimension 0.
- l. Any two bases of a subspace W of $\mathbb{R}^n$ contain the same number of vectors.
- m. Every independent subset of $\mathbb{R}^n$ is a subset of every basis for $\mathbb{R}^n$.
- n. Every independent subset of $\mathbb{R}^n$ is a subset of some basis for $\mathbb{R}^n$.
29. Let u and v be two different vectors in $\mathbb{R}^n$. Prove that $\{u, v\}$ is linearly dependent if and only if one of the vectors is a multiple of the other.
30. Let v_1, v_2, v_3 be independent vectors in $\mathbb{R}^n$. Prove that $w_1 = 3v_1$, $w_2 = 2v_1 - v_2$, and $w_3 = v_1 + v_3$ are also independent.
31. Let v_1, v_2, v_3 be any vectors in $\mathbb{R}^n$. Prove that $w_1 = 2v_1 + 3v_2$, $w_2 = v_2 - 2v_3$, and $w_3 = -v_1 - 3v_3$ are dependent.
32. Find all scalars s , if any exist, such that $[1, 0, 1], [2, s, 3], [2, 3, 1]$ are independent.
33. Find all scalars s , if any exist, such that $[1, 0, 1], [2, s, 3], [1, -s, 0]$ are independent.

34. Let $\mathbf{v}$ and $\mathbf{w}$ be independent column vectors in $\mathbb{R}^3$, and let A be an invertible 3×3 matrix. Prove that the vectors $A\mathbf{v}$ and $A\mathbf{w}$ are independent.
35. Give an example showing that the conclusion of the preceding exercise need not hold if A is nonzero but singular. Can you also find specific independent vectors $\mathbf{v}$ and $\mathbf{w}$ and a singular matrix A such that $A\mathbf{v}$ and $A\mathbf{w}$ are still independent?
36. Let $\mathbf{v}$ and $\mathbf{w}$ be column vectors in $\mathbb{R}^n$, and let A be an $n \times n$ matrix. Prove that, if $A\mathbf{v}$ and $A\mathbf{w}$ are independent, $\mathbf{v}$ and $\mathbf{w}$ are independent.
37. Generalizing Exercise 34, let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ be independent column vectors in $\mathbb{R}^n$, and let C be an invertible $n \times n$ matrix. Prove that the vectors $C\mathbf{v}_1, C\mathbf{v}_2, \dots, C\mathbf{v}_k$ are independent.
38. Prove that if W is a subspace of $\mathbb{R}^n$ and $\dim(W) = n$, then $W = \mathbb{R}^n$.
-  In Exercises 39–42, use LINTeK to find a basis for the space spanned by the given vectors in $\mathbb{R}^n$.
39. $\mathbf{v}_1 = [5, 4, 3], \quad \mathbf{v}_4 = [6, 1, 4],$
 $\mathbf{v}_2 = [2, 1, 6], \quad \mathbf{v}_5 = [1, 1, 1]$
 $\mathbf{v}_3 = [4, 5, -12],$
40. $\mathbf{a}_1 = [0, 1, 1, 2], \quad \mathbf{a}_4 = [-1, 4, 6, 11],$
 $\mathbf{a}_2 = [-3, -2, 4, 5], \quad \mathbf{a}_5 = [1, 1, 1, 3],$
 $\mathbf{a}_3 = [1, 2, 0, 1], \quad \mathbf{a}_6 = [3, 7, 3, 9]$
41. $\mathbf{u}_1 = [3, 1, 2, 4, 1],$
 $\mathbf{u}_2 = [3, -2, 6, 7, -3],$
 $\mathbf{u}_3 = [3, 4, -2, 1, 5],$
 $\mathbf{u}_4 = [1, 2, 3, 2, 1],$
 $\mathbf{u}_5 = [7, 1, 11, 13, -1],$
 $\mathbf{u}_6 = [2, -1, 2, 3, 1]$
42. $\mathbf{w}_1 = [2, -1, 3, 4, 1, 2],$
 $\mathbf{w}_2 = [-2, 5, 3, -2, 1, -4],$
 $\mathbf{w}_3 = [2, 4, 6, 5, 2, 1],$
 $\mathbf{w}_4 = [1, -1, 1, -1, 2, 2],$
 $\mathbf{w}_5 = [1, 8, 10, 2, 5, -1],$
 $\mathbf{w}_6 = [3, 0, 0, 2, 1, 5]$

MATLAB

Access MATLAB and work the indicated exercise. If the data files for the text are available, enter `fb2s1` for the vector data. Otherwise, enter the vector data by hand.

- M1. Exercise 39
 M2. Exercise 40
 M3. Exercise 41
 M4. Exercise 42

2.2

THE RANK OF A MATRIX

In Section 1.6 we discussed three subspaces associated with an $m \times n$ matrix A : its *column space* in $\mathbb{R}^m$, its *row space* in $\mathbb{R}^n$, and its *nullspace* (solution space of $A\mathbf{x} = \mathbf{0}$) in $\mathbb{R}^n$. In this section we consider how the dimensions of these subspaces are related.

We can find the dimension of the column space of A by row-reducing A to row-echelon form H . This dimension is the number of columns of H having pivots.

Turning to the row space, note that interchange of rows does not change the row space, and neither does multiplication of a row by a nonzero scalar. If we multiply the i th row vector $\mathbf{v}_i$ by a scalar r and add it to the k th row vector $\mathbf{v}_k$, then the new k th row vector is $r\mathbf{v}_i + \mathbf{v}_k$, which is still in the row space of A .

because it is a linear combination of rows of A . But the original row vector $\mathbf{v}_k$ is also in the row space of the new matrix, because it is equal to $(r\mathbf{v}_i + \mathbf{v}_k) + (-r)\mathbf{v}_i$. Thus row addition also does not change the row space of a matrix.

Suppose that the reduced row-echelon form of a matrix A is

$$H = \begin{bmatrix} 1 & 0 & 2 & 0 & 5 \\ 0 & 1 & -3 & 0 & 9 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The configuration of the three nonzero row vectors in their 1st, 2nd, and 4th components is the configuration of the row vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ in $\mathbb{R}^3$, and ensures that the first three row vectors of H are independent. In this way we see that the dimension of the row space of any matrix is the number of nonzero rows in its reduced row-echelon form, or just in its row-echelon form. But this is also the number of pivots in the matrix. Thus the dimension of the column space of A must be equal to the dimension of its row space. This common dimension of the row space and the column space is the **rank** of A , denoted by $\text{rank}(A)$. These arguments generalize to prove the following theorem.

THEOREM 2.4 Row Rank Equals Column Rank

Let A be an $m \times n$ matrix. The dimension of the row space of A is equal to the dimension of the column space of A . The common dimension, the **rank** of A , is the number of pivots in a row-echelon form of A .

We know that a basis for the column space of A consists of the columns of A giving rise to pivots in a row-echelon form of A . We saw how to find a basis for the nullspace of A in Section 1.6. We would like to be able to find a basis for the row space. We could work with the transpose of A , but this would require

HISTORICAL NOTE THE RANK OF A MATRIX was defined in 1879 by Georg Frobenius (1849–1917) as follows: If all determinants of the $(r+1)$ st degree vanish, but not all of the r th degree, then r is the rank of the matrix. Frobenius used this concept to deal with the questions of canonical forms for certain matrices of integers and with the solutions of certain systems of linear congruences.

The nullity was defined by James Sylvester in 1884 for square matrices as follows: The nullity of an $n \times n$ matrix is i if every minor (determinant) of order $n-i+1$ (and therefore of every higher order) equals 0 and i is the largest such number for which this is true. Sylvester was interested here, as in much of his mathematical career, in discovering invariants—properties of particular mathematical objects that do not change under specified types of transformations. He proceeded to prove what he called one of the cardinal laws in the theory of matrices, that the nullity of the product of two matrices is not less than the nullity of any factor or greater than the sum of the nullities of the factors.

another reduction to row-echelon form. In fact, because the elementary row operations do not change the row space of A , we simply can take as a basis for the row space of A the nonzero rows in a row-echelon form. We summarize in a box.

Finding Bases for Spaces Associated with a Matrix

Let A be an $m \times n$ matrix with row-echelon form H .

1. For a basis of the row space of A , use the nonzero rows of H .
2. For a basis of the column space of A , use the columns of A corresponding to the columns of H containing pivots.
3. For a basis of the nullspace of A , use H and back substitution to solve $Hx = 0$ in the usual way (see Example 3, Section 1.6).

EXAMPLE 1 Find the rank, a basis for the row space, a basis for the column space, and a basis for the nullspace of the matrix

$$A = \begin{bmatrix} 1 & 3 & 0 & -1 & 2 \\ 0 & -2 & 4 & -2 & 0 \\ 3 & 11 & -4 & -1 & 6 \\ 2 & 5 & 3 & -4 & 0 \end{bmatrix}.$$

SOLUTION We reduce A all the way to reduced row-echelon form, because we also want to find a basis for the nullspace of A . We obtain

$$A = \begin{bmatrix} 1 & 3 & 0 & -1 & 2 \\ 0 & -2 & 4 & -2 & 0 \\ 3 & 11 & -4 & -1 & 6 \\ 2 & 5 & 3 & -4 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 0 & -1 & 2 \\ 0 & -2 & 4 & -2 & 0 \\ 0 & 2 & -4 & 2 & 0 \\ 0 & -1 & 3 & -2 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 6 & -4 & 2 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & -4 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & 0 & 0 & 2 & 26 \\ 0 & 1 & 0 & -1 & -8 \\ 0 & 0 & 1 & -1 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = H.$$

Because the reduced form H contains three pivots, we see that $\text{rank}(A) = 3$.

As a basis for the row space of A , we take the nonzero row vectors of H , obtaining

$$\{[1, 0, 0, 2, 26], [0, 1, 0, -1, -8], [0, 0, 1, -1, -4]\}.$$

Notice that the next to the last matrix in the reduction shows that the first three row vectors of A are dependent, so we must not take them as a basis for the row space.

Now the columns of A in which pivots appear in H form a basis for the column space, and from H we see that the solution of $Ax = 0$ is

$$x = \begin{bmatrix} -2r - 26s \\ r + 8s \\ r + 4s \\ r \\ s \end{bmatrix} = r \begin{bmatrix} -2 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -26 \\ 8 \\ 4 \\ 0 \\ 1 \end{bmatrix}. \text{ Thus we have the following bases:}$$

$$\text{Column space: } \left\{ \begin{bmatrix} 1 \\ 0 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ 11 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ -4 \\ 3 \end{bmatrix} \right\} \quad \text{Null space: } \left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -26 \\ 8 \\ 4 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The Rank Equation

Let A be an $m \times n$ matrix. Recall that the nullspace of A —that is, the solution set of $Ax = 0$, has a basis with as many vectors as the number of free scalar variables, like r and s , appearing in the solution vector above. Because we have one free scalar variable for each column without a pivot in a row-echelon form of A , we see that the dimension of the nullspace of A is the number of columns of A that do not contain a pivot. This dimension is called the **nullity** of A , and is denoted by $\text{nullity}(A)$. Because the number of columns that do have a pivot is the dimension, $\text{rank}(A)$, of the column space of A , we see that

$$\text{rank}(A) + \text{nullity}(A) = n, \quad \text{Rank equation}$$

where n is the number of columns of A . This equation turns out to be very useful. We summarize this equation, and the method for computing the numbers it involves, in a theorem.

THEOREM 2.5 Rank Equation

Let A be an $m \times n$ matrix with row-echelon form H . Then:

1. $\text{nullity}(A) = (\text{Number of free variables in the solution space of } Ax = 0) = (\text{Number of pivot-free columns in } H);$
2. $\text{rank}(A) = (\text{Number of pivots in } H);$
3. (Rank equation) $\text{rank}(A) + \text{nullity}(A) = (\text{Number of columns of } A).$

Because $\text{nullity}(A)$ is defined as the number of vectors in a basis of the nullspace of A , the invariance of dimension shows that the number of free variables obtained in the solution of a linear system $Ax = b$ is independent of the steps in the row reduction to echelon form, as we asserted in Section 1.4.

EXAMPLE 2 Illustrate the rank equation for the matrix A in Example 1.

SOLUTION The matrix A in Example 1 has $n = 5$ columns, and we saw that $\text{rank}(A) = 3$ and $\text{nullity}(A) = 2$. Thus the rank equation is $3 + 2 = 5$. ■

Our work has given us still another criterion for the invertibility of a square matrix.

THEOREM 2.6 An Invertibility Criterion

An $n \times n$ matrix A is invertible if and only if $\text{rank}(A) = n$.

SUMMARY

1. Let A be an $m \times n$ matrix. The dimension of the row space of A is equal to the dimension of the column space of A , and is called the *rank* of A , denoted by $\text{rank}(A)$. The rank of A is equal to the number of pivots in a row-echelon form H of A . The *nullity* of A , denoted by $\text{nullity}(A)$, is the dimension of the nullspace of A —that is, of the solution set of $Ax = 0$.
2. Bases for the row space, the column space, and the nullspace of a matrix A can be found as described in a box in the text.
3. (*Rank Equation*) For an $m \times n$ matrix A , we have

$$\text{rank}(A) + \text{nullity}(A) = n.$$

EXERCISES

For the matrices in Exercises 1–6, find (a) the rank of the matrix, (b) a basis for the row space, (c) a basis for the column space, and (d) a basis for the nullspace.

$$1. \begin{bmatrix} 2 & 0 & -3 & 1 \\ 3 & 4 & 2 & 2 \end{bmatrix} \quad 2. \begin{bmatrix} 5 & -1 & 0 & 2 \\ 1 & 2 & 1 & 0 \\ 3 & 1 & -2 & 4 \\ 0 & 4 & -1 & 2 \end{bmatrix}$$

$$5. \begin{bmatrix} 0 & 1 & 2 & 1 \\ 2 & 1 & 0 & 2 \\ 0 & 2 & 1 & 1 \end{bmatrix}$$

$$6. \begin{bmatrix} 0 & 2 & 3 & 1 \\ -4 & 4 & 1 & 4 \\ 3 & 3 & 2 & 0 \\ -4 & 0 & 1 & 2 \end{bmatrix}$$

In Exercises 7–10, determine whether the given matrix is invertible, by finding its rank.

$$7. \begin{bmatrix} 0 & -9 & -9 & 2 \\ 1 & 2 & 1 & 1 \\ 4 & 1 & -3 & 4 \\ 1 & 3 & 2 & 0 \end{bmatrix}$$

$$8. \begin{bmatrix} 2 & 3 & 1 \\ 4 & -1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$$

$$3. \begin{bmatrix} 0 & 6 & 6 & 3 \\ 1 & 2 & 1 & 1 \\ 4 & 1 & -3 & 4 \\ 1 & 3 & 2 & 0 \end{bmatrix} \quad 4. \begin{bmatrix} 3 & 1 & 4 & 2 \\ -1 & 0 & -1 & 0 \\ 2 & 1 & 0 & 1 \\ 1 & 0 & -1 & 1 \end{bmatrix}$$

$$9. \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 4 \\ 2 & 4 & 0 \end{bmatrix}$$

$$10. \begin{bmatrix} 3 & 0 & -1 & 2 \\ 4 & 2 & 1 & 8 \\ 1 & 4 & 0 & 1 \\ 2 & 6 & -3 & 1 \end{bmatrix}$$

11. Mark each of the following True or False.
- a. The number of independent row vectors in a matrix is the same as the number of independent column vectors.
 - b. If H is a row-echelon form of a matrix A , then the nonzero column vectors in H form a basis for the column space of A .
 - c. If H is a row-echelon form of a matrix A , then the nonzero row vectors in H are a basis for the row space of A .
 - d. If an $n \times n$ matrix A is invertible, then $\text{rank}(A) = n$.
 - e. For every matrix A , we have $\text{rank}(A) > 0$.
 - f. For all positive integers m and n , the rank of an $m \times n$ matrix might be any number from 0 to the maximum of m and n .
 - g. For all positive integers m and n , the rank of an $m \times n$ matrix might be any number from 0 to the minimum of m and n .
 - h. For all positive integers m and n , the nullity of an $m \times n$ matrix might be any number from 0 to n .
 - i. For all positive integers m and n , the nullity of an $m \times n$ matrix might be any number from 0 to m .
 - j. For all positive integers m and n , with $m \geq n$, the nullity of an $m \times n$ matrix might be any number from 0 to n .
12. Prove that, if A is a square matrix, the nullity of A is the same as the nullity of A^T .
13. Let A be an $m \times n$ matrix, and let $\mathbf{b}$ be an $n \times 1$ vector. Prove that the system of equations $A\mathbf{x} = \mathbf{b}$ has a solution for $\mathbf{x}$ if and only if $\text{rank}(A) = \text{rank}(A | \mathbf{b})$, where $\text{rank}(A | \mathbf{b})$ represents the rank of the associated augmented matrix $[A | \mathbf{b}]$ of the system.

In Exercises 14–16, let A and C be matrices such that the product AC is defined.

14. Prove that the column space of AC is contained in the column space of A .
15. Is it true that the column space of AC is contained in the column space of C ? Explain.
16. State the analogue of Exercise 14 concerning the row spaces of A and C .

17. Give an example of a 3×3 matrix A such that $\text{rank}(A) = 2$ and $\text{rank}(A^3) = 0$.

In Exercises 18–20, let A and C be matrices such that the product AC is defined.

18. Prove that $\text{rank}(AC) \leq \text{rank}(A)$.
19. Give an example where $\text{rank}(AC) < \text{rank}(A)$.
20. Is it true that $\text{rank}(AC) \leq \text{rank}(C)$? Explain.

It can be shown that $\text{rank}(A^T A) = \text{rank}(A)$ (see Theorem 6.10). Use this result in Exercises 21–23.

21. Let A be an $m \times n$ matrix. Prove that $\text{rank}(A(A^T)) = \text{rank}(A)$.
22. If $\mathbf{a}$ is an $n \times 1$ vector and $\mathbf{b}$ is a $1 \times m$ vector, prove that $\mathbf{ab}$ is an $n \times m$ matrix of rank at most one.
23. Let A be an $m \times n$ matrix. Prove that the column space and row space of $(A^T)A$ are the same.
24. Suppose that you are using computer software, such as LINTEK or MATLAB, that will compute and print the reduced row-echelon form of a matrix but does not indicate any row interchanges it may have made. How can you determine what rows of the original matrix form a basis for the row space?

In Exercises 25 and 26, use LINTEK or MATLAB to request a row reduction of the matrix, without seeing intermediate steps. Load data files as usual if they are available. (a) Give the rank of the matrix, and (b) use the software as suggested in Exercise 24 to find the lowest numbered rows, in consecutive order, of the given matrix that form a basis for its row space.

$$25. A = \begin{bmatrix} 2 & -3 & 0 & 1 & 4 \\ 1 & 4 & -6 & 3 & -2 \\ 0 & 11 & -12 & 5 & -8 \\ 4 & -1 & 5 & 3 & 7 \end{bmatrix}$$

$$26. B = \begin{bmatrix} -1 & 1 & 3 & -6 & 8 & -2 \\ -3 & 5 & 3 & 1 & 4 & 8 \\ 1 & -3 & 3 & -13 & 12 & -12 \\ 0 & 2 & -6 & 19 & -20 & 14 \\ 5 & 13 & -21 & 3 & 11 & 6 \end{bmatrix}$$

2.3

LINEAR TRANSFORMATIONS OF EUCLIDEAN SPACES

When we introduced the notation $Ax = \mathbf{b}$ and indicated why it is one of the most useful notations in mathematics, we mentioned that we would see that we could regard A as a function and view $Ax = \mathbf{b}$ as meaning that the function maps the vector $\mathbf{x}$ to the vector $\mathbf{b}$. If A is an $m \times n$ matrix and the product Ax is defined, then $\mathbf{x} \in \mathbb{R}^n$ can be viewed as the input variable and $\mathbf{b} \in \mathbb{R}^m$ can be viewed as the output variable.

Functions are used throughout mathematics to study the structures of sets and relationships between sets. You are familiar with the notation $y = f(x)$, where f is a function that acts on numbers, signified by the input variable x , and produces numbers signified by the output variable y . In linear algebra, we are interested in functions $\mathbf{y} = f(\mathbf{x})$, where f acts on vectors, signified by the input variable $\mathbf{x}$, and produces vectors signified by the output variable $\mathbf{y}$.

In general, a function $f: X \rightarrow Y$ is a rule that associates with each x in the set X an element $y = f(x)$ in Y . We say that f **maps the set** X into the set Y and **maps the element** x to the element y . The set X is the **domain** of f and the set Y is called the **codomain**. To describe a function, we must give its domain and codomain, and then we must specify the action of the function on each element of its domain. For any subset H of X , we let $f[H] = \{f(h) \mid h \in H\}$; the set $f[H]$ is called the **image** of H under f . The image of the domain of f is the **range** of f . Likewise, for a subset K of Y , the set $f^{-1}[K] = \{x \in X \mid f(x) \in K\}$ is the **inverse image** of K under f . This is illustrated in Figure 2.2. For example, if $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = x^2$, then $f[\{1, 2, 3\}] = \{1, 4, 9\}$ and $f^{-1}[\{1, 4, 9\}] = \{-1, 1, -2, 2, -3, 3\}$. In this section, we study functions known as *linear transformations* T that have as domain $\mathbb{R}^n$ and as codomain $\mathbb{R}^m$, as depicted in Figure 2.3.

The Notion of a Linear Transformation

Notice that for an $m \times n$ matrix A , the function mapping $\mathbf{x} \in \mathbb{R}^n$ into $A\mathbf{x}$ in $\mathbb{R}^m$ satisfies the two conditions in the following definition (see Example 3).

DEFINITION 2.3 Linear Transformation

A function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a **linear transformation** if it satisfies two conditions:

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ **Preservation of addition**
2. $T(r\mathbf{u}) = rT(\mathbf{u})$ **Preservation of scalar multiplication**

for all vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ and for all scalars r .

From properties 1 and 2, it follows that $T(r\mathbf{u} + s\mathbf{v}) = rT(\mathbf{u}) + sT(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and all scalars r and s . (See Exercise 32.) In fact, this equation can be

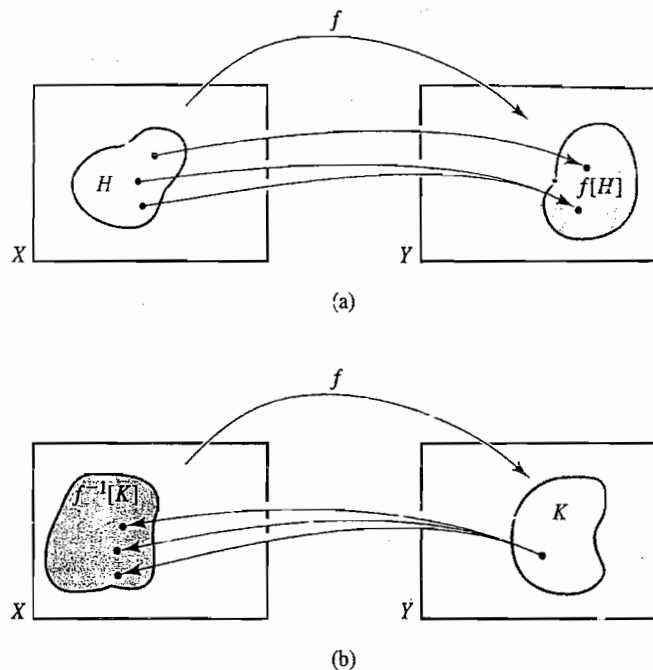


FIGURE 2.2

(a) The image of H under f ; (b) the inverse image of K under f .

extended to any number of summands by induction—that is, for $v_1, v_2, \dots, v_k$ in $\mathbb{R}^n$ and scalars $r_1, r_2, \dots, r_k$, we have

$$T(r_1v_1 + r_2v_2 + \dots + r_kv_k) = r_1T(v_1) + r_2T(v_2) + \dots + r_kT(v_k). \quad (1)$$

Equation (1) is often expressed verbally as follows:

Linear transformations preserve linear combinations.

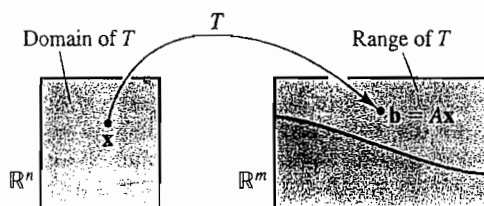


FIGURE 2.3

The linear transformation $T(x) = Ax$.

We claim that if $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation, then T maps the zero vector of $\mathbb{R}^n$ to the zero vector of $\mathbb{R}^m$. Just observe that scalar multiplication of any vector by the zero scalar gives the zero vector, and property 2 of the definition shows that

$$T(\mathbf{0}) = T(0\mathbf{0}) = 0T(\mathbf{0}) = \mathbf{0}.$$

EXAMPLE 1 Show from Definition 2.3 that the function $T: \mathbb{R} \rightarrow \mathbb{R}$ defined by $T(x) = \sin x$ is not a linear transformation.

SOLUTION We know that

$$\sin\left(\frac{\pi}{4} + \frac{\pi}{4}\right) \neq \sin\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right),$$

because $\sin(\pi/4 + \pi/4) = \sin(\pi/2) = 1$, but we have $\sin(\pi/4) + \sin(\pi/4) = 1/\sqrt{2} + 1/\sqrt{2} = 2/\sqrt{2}$. Thus, $\sin x$ is not a linear transformation, because it does not preserve addition. ■

EXAMPLE 2 Determine whether $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T([x_1, x_2]) = [x_2, x_1 - x_2, 2x_1 + x_2]$ is a linear transformation.

SOLUTION To test for preservation of addition, we let $\mathbf{u} = [u_1, u_2]$ and $\mathbf{v} = [v_1, v_2]$, and compute

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T([u_1 + v_1, u_2 + v_2]) \\ &= [u_2 + v_2, u_1 + v_1 - u_2 - v_2, 2u_1 + 2v_1 + u_2 + v_2] \\ &= [u_2, u_1 - u_2, 2u_1 + u_2] + [v_2, v_1 - v_2, 2v_1 + v_2] \\ &= T(\mathbf{u}) + T(\mathbf{v}), \end{aligned}$$

and so vector addition is preserved. To test for preservation of scalar multiplication, we compute

$$\begin{aligned} T(r\mathbf{u}) &= T([ru_1, ru_2]) = [ru_2, ru_1 - ru_2, 2ru_1 + ru_2] \\ &= r[u_2, u_1 - u_2, 2u_1 + u_2] \\ &= rT(\mathbf{u}). \end{aligned}$$

Thus, scalar multiplication is also preserved, and so T is a linear transformation. ■

EXAMPLE 3 Let A be an $m \times n$ matrix, and let $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be defined by $T_A(\mathbf{x}) = A\mathbf{x}$ for each column vector $\mathbf{x} \in \mathbb{R}^n$. Show that T_A is a linear transformation.

SOLUTION This follows from the distributive and scalars-pull-through properties of matrix multiplication stated in Section 1.3—namely, for any vectors $\mathbf{u}$ and $\mathbf{v}$ and for any scalar r , we have

$$T_A(\mathbf{u} + \mathbf{v}) = A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = T_A(\mathbf{u}) + T_A(\mathbf{v})$$

and

$$T_A(r\mathbf{u}) = A(r\mathbf{u}) = r(A\mathbf{u}) = rT_A(\mathbf{u}).$$

These are precisely the conditions for a linear transformation given in Definition 2.3. ■

Looking back at Example 2 we see that, in column-vector notation, the transformation there appears as

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_2 \\ x_1 - x_2 \\ 2x_1 + x_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Based on Example 3, we can conclude that T is a linear transformation, obviating the need to check linearity directly as we did in Example 2. In a moment, we will show that *every* linear transformation of $\mathbb{R}^n$ into $\mathbb{R}^m$ has the form $T(\mathbf{x}) = A\mathbf{x}$ for some $m \times n$ matrix A . This is especially easy to see for linear transformations of $\mathbb{R}$ into $\mathbb{R}$.

EXAMPLE 4 Determine all linear transformations of $\mathbb{R}$ into $\mathbb{R}$.

SOLUTION Let $T: \mathbb{R} \rightarrow \mathbb{R}$ be a linear transformation. Each element of $\mathbb{R}$ can be viewed either as a vector or as a scalar. Let $a = T(1)$. Applying Property 2 of Definition 2.3 with $\mathbf{u} = 1$ and $r = x$, we obtain

$$T(x) = T(x(1)) = xT(1) = xa = ax.$$

Identifying a with the 1×1 matrix A having a as its sole entry, we see that we have $T(\mathbf{x}) = A\mathbf{x}$, and we know this transformation satisfies properties 1 and 2 in Definition 2.3. From a geometric viewpoint, we see that the *linear* transformations of $\mathbb{R}$ into $\mathbb{R}$ can be described as precisely those functions whose graphs are *lines through the origin*. ■

Example 4 shows that a linear transformation of $\mathbb{R}$ into $\mathbb{R}$ is completely determined as soon as $T(1)$ is known. More generally, a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is uniquely determined by its values on any basis for $\mathbb{R}^n$, as we now show.

THEOREM 2.7 Bases and Linear Transformations

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis for $\mathbb{R}^n$. For any vector $\mathbf{v}$ in $\mathbb{R}^n$, the vector $T(\mathbf{v})$ is uniquely determined by the vectors $T(\mathbf{b}_1), T(\mathbf{b}_2), \dots, T(\mathbf{b}_n)$.

PROOF Let $\mathbf{v}$ be any vector in $\mathbb{R}^n$. We know that because B is a basis, there exist *unique* scalars $r_1, r_2, \dots, r_n$ such that

$$\mathbf{v} = r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \cdots + r_n\mathbf{b}_n.$$

Using Eq. (1), we see that

$$T(\mathbf{v}) = T(r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \cdots + r_n\mathbf{b}_n) = r_1T(\mathbf{b}_1) + r_2T(\mathbf{b}_2) + \cdots + r_nT(\mathbf{b}_n).$$

Because the coefficients r_i are *uniquely* determined by $\mathbf{v}$, it follows that $T(\mathbf{v})$ is completely determined by the vectors $T(\mathbf{b}_i)$ for $i = 1, 2, \dots, n$. $\blacktriangle$

Theorem 2.7 shows that if two linear transformations have the same value at each basis vector $\mathbf{b}_i$, then the two transformations have the same value at each vector in $\mathbb{R}^n$, and thus they are the same transformation.

COROLLARY Standard Matrix Representation of a Linear Transformation

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and let A be the $m \times n$ matrix whose j th column vector is $T(\mathbf{e}_j)$, which we denote symbolically as

$$A = \begin{bmatrix} | & | & & | \\ T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \\ | & | & & | \end{bmatrix}. \quad (2)$$

Then $T(\mathbf{x}) = A\mathbf{x}$ for each column vector $\mathbf{x} \in \mathbb{R}^n$.

PROOF Recall that for any matrix A , $A\mathbf{e}_j$ is the j th column of A . This shows at once that if A is the matrix described in Eq. (2), then $A\mathbf{e}_j = T(\mathbf{e}_j)$, and so T and the linear transformation T_A given by $T_A(\mathbf{x}) = A\mathbf{x}$ agree on the standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n$. By Theorem 2.7, and the comment following this theorem, we know that then $T(\mathbf{x}) = T_A(\mathbf{x})$ for every $\mathbf{x} \in \mathbb{R}^n$ —that is, $T(\mathbf{x}) = A\mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^n$. $\blacktriangle$

The matrix A in Eq. (2) is the **standard matrix representation** of the linear transformation T .

EXAMPLE 5 Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the linear transformation such that

$$T(\mathbf{e}_1) = [2, 1, 4] \quad \text{and} \quad T(\mathbf{e}_2) = [3, 0, -2].$$

Find the standard matrix representation A of T and find a formula for $T([x_1, x_2])$.

SOLUTION Equation (2) for the standard matrix representation shows that

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 0 \\ 4 & -2 \end{bmatrix}, \text{ so } T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2 & 3 \\ 1 & 0 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + 3x_2 \\ x_1 \\ 4x_1 - 2x_2 \end{bmatrix}.$$

In row-vector notation, we have the formula

$$T([x_1, x_2]) = [2x_1 + 3x_2, x_1, 4x_1 - 2x_2].$$

EXAMPLE 6 Find the standard matrix representation of the linear transformation $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ where

$$T([x_1, x_2, x_3, x_4]) = [x_2 - 3x_3, 2x_1 - x_2 + 3x_4, 8x_1 - 4x_2 + 3x_3 - x_4]. \quad (3)$$

SOLUTION We compute

$$T(\mathbf{e}_1) = T([1, 0, 0, 0]) = [0, 2, 8], \quad T(\mathbf{e}_2) = T([0, 1, 0, 0]) = [1, -1, -4]$$

$$T(\mathbf{e}_3) = T([0, 0, 1, 0]) = [-3, 0, 3], \quad T(\mathbf{e}_4) = T([0, 0, 0, 1]) = [0, 3, -1].$$

Using Eq. (2), we find that

$$A = \begin{bmatrix} 0 & 1 & -3 & 0 \\ 2 & -1 & 0 & 3 \\ 8 & -4 & 3 & -1 \end{bmatrix}.$$

Perhaps you noticed in Example 6 that the first row of the matrix A consists of the coefficients of x_1, x_2, x_3 , and x_4 in the first component $x_2 - 3x_3$ of $T([x_1, x_2, x_3, x_4])$. The second and third rows can be found similarly. If Eq. (3) is written in column-vector form, the matrix A jumps out at you immediately. Try it! This is often a fast way to write down the standard matrix representation when the transformation is described by a formula as in Example 6. Be sure to remember, however, the Eq. (2) formulation for the standard matrix representation.

We give another example indicating how a linear transformation is determined, as in the proof of Theorem 2.7, if we know its values on a basis for its domain. Note that the vectors $\mathbf{u} = [-1, 2]$ and $\mathbf{v} = [3, -5]$ are two nonparallel vectors in the plane, and form a basis for $\mathbb{R}^2$.

EXAMPLE 7 Let $\mathbf{u} = [-1, 2]$ and $\mathbf{v} = [3, -5]$ be in $\mathbb{R}^2$, and let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation such that $T(\mathbf{u}) = [-2, 1, 0]$ and $T(\mathbf{v}) = [5, -7, 1]$. Find the standard matrix representation A of T and compute $T([-4, 3])$.

SOLUTION To find the standard matrix representation of T , we need to find $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$ for $\mathbf{e}_1, \mathbf{e}_2 \in \mathbb{R}^2$. Following the argument in the proof of Theorem 2.7, we express $\mathbf{e}_1$ and $\mathbf{e}_2$ as linear combinations of the basis vectors $\mathbf{u}$ and $\mathbf{v}$ for $\mathbb{R}^2$, where we know the action of T . To express $\mathbf{e}_1$ and $\mathbf{e}_2$ as linear combinations of $\mathbf{u}$ and $\mathbf{v}$, we solve the two linear systems $A\mathbf{x} = \mathbf{e}_1$ and $A\mathbf{x} = \mathbf{e}_2$, where the coefficient matrix A has $\mathbf{u}$ and $\mathbf{v}$ as its column vectors. Because both systems have the same coefficient matrix, we can solve them both at once as follows:

$$\left[\begin{array}{cc|cc} -1 & 3 & 1 & 0 \\ 2 & -5 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & -3 & -1 & 0 \\ 0 & 1 & 2 & 1 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 5 & 3 \\ 0 & 1 & 2 & 1 \end{array} \right].$$

We see that $\mathbf{e}_1 = 5\mathbf{u} + 2\mathbf{v}$ and $\mathbf{e}_2 = 3\mathbf{u} + \mathbf{v}$. Using the linearity properties,

$$\begin{aligned} T(\mathbf{e}_1) &= T(5\mathbf{u} + 2\mathbf{v}) = 5T(\mathbf{u}) + 2T(\mathbf{v}) = 5[-2, 1, 0] + 2[5, -7, 1] \\ &= [0, -9, 2] \end{aligned}$$

and

$$\begin{aligned} T(\mathbf{e}_2) &= T(3\mathbf{u} + \mathbf{v}) = 3T(\mathbf{u}) + T(\mathbf{v}) = 3[-2, 1, 0] + [5, -7, 1] \\ &= [-1, -4, 1]. \end{aligned}$$

The standard matrix representation A of T and $T([-4, 3])$ are

$$A = \begin{bmatrix} 0 & -1 \\ -9 & -4 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad T\left(\begin{bmatrix} -4 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 0 & -1 \\ -9 & -4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -4 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 24 \\ -5 \end{bmatrix}.$$

Some Terminology of Linear Transformations

The matrix representation A of a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a great help in working with T . Let us use column-vector notation. Suppose, for example, we want to find the range of T —that is, the set of all elements of $\mathbb{R}^m$ that are equal to $T(\mathbf{x})$ for some $\mathbf{x} \in \mathbb{R}^n$. Recall that we use the notation $T[\mathbb{R}^n]$ to denote the set of all these elements, so that $T[\mathbb{R}^n] = \{T(\mathbf{x}) \mid \mathbf{x} \in \mathbb{R}^n\}$. Because $T(\mathbf{x}) = A\mathbf{x}$, we have $T[\mathbb{R}^n] = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n\}$. Now $A\mathbf{x}$ is a linear combination of the column vectors of A where the coefficient of the j th column of A in the linear combination is x_j , the j th component of the vector $\mathbf{x}$. Thus the range of T is precisely the column space of A .

For another illustration, finding all $\mathbf{x}$ such that $T(\mathbf{x}) = \mathbf{0}$ amounts to solving the linear system $A\mathbf{x} = \mathbf{0}$. We know that the solution of this homogeneous linear system is a subspace of $\mathbb{R}^n$, called the *nullspace* of the matrix A . This nullspace is often called the **kernel** of T as well as the **nullspace** of T , and is denoted $\ker(T)$.

Let W be a subspace of $\mathbb{R}^n$. Then $W = \text{sp}(\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k)$ where $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ is a basis for W . Because T preserves linear combinations, we have

$$T(r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \dots + r_k\mathbf{b}_k) = r_1T(\mathbf{b}_1) + r_2T(\mathbf{b}_2) + \dots + r_kT(\mathbf{b}_k).$$

This shows that $T[W] = \text{sp}(T(\mathbf{b}_1), T(\mathbf{b}_2), \dots, T(\mathbf{b}_k))$, which we know is a subspace of $\mathbb{R}^m$.

We summarize the three preceding paragraphs in a box.

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation with standard matrix representation A .

1. The *range* $T[\mathbb{R}^n]$ of T is the column space of A in $\mathbb{R}^m$.
2. The *kernel* of T is the nullspace of A and is denoted $\ker(T)$.
3. If W is a subspace of $\mathbb{R}^n$, then $T[W]$ is a subspace of $\mathbb{R}^m$; that is, T preserves subspaces.